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ក្រសួងអប់រំ យុវជន និងកីឡា
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**“CAMBODIA DAILY CONSUMER PRICE INDEX (CPI) PIPELINE:
HIGH-FREQUENCY WEB SCRAPING, SEMANTIC ENTITY MATCHING,
COICOP CLASSIFICATION, AND REAL-TIME INFLATION NOWCASTING”**

at

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សន្ទស្សន៍ថ្លៃទំនិញប្រើប្រាស់ (CPI) គឺជាសូចនាករគន្លឹះក្នុងការវាស់វែងអតិផរណា និងតាមដានកម្លាំងទិញជាក់ស្តែងរបស់ប្រជាពលរដ្ឋ ប៉ុន្តែការស្ទង់មតិបែបប្រពៃណីនៅកម្ពុជាមានគម្លាតពេលវេលាយឺតយ៉ាវពី ៣០ ទៅ ៦០ ថ្ងៃ និងចំណាយថវិកាច្រើន។ ដើម្បីដោះស្រាយបញ្ហានេះ គម្រោងស្រាវជ្រាវនេះបានបង្កើតប្រព័ន្ធវាស់វែងសន្ទស្សន៍ថ្លៃទំនិញប្រើប្រាស់ប្រចាំថ្ងៃនៅកម្ពុជា (**Cambodia Daily CPI Medallion Pipeline**) ដោយប្រមូលទិន្នន័យថ្លៃទំនិញស្វ័យប្រវត្តិជារៀងរាល់ព្រឹកនៅម៉ោង ០៨:០០ ព្រឹក (ICT) ពី ២៥ ប្រភពអនឡាញ គ្របដណ្តប់លើទំហំ ១២ ជំពូក UN COICOP ក្នុងរាជធានីភ្នំពេញ តាមរយៈស្ថាបត្យកម្ម PostgreSQL 16 Medallion (Bronze, Silver, Gold)។

នៅក្នុងស្រទាប់ Silver ប្រព័ន្ធអនុវត្តការផ្គុំផ្គង់អត្តសញ្ញាណទំនិញតាមរយៈ Vector Embeddings (gemini-embedding-2) និងវិធានការពារលក្ខណៈបច្ចេកទេសចំនួន ៧។ ចំណែកការចាត់ថ្នាក់ទំនិញចូលជំពូក ៤ ខ្ទង់ (DD.G.C) ត្រូវបានអនុវត្តតាមយន្តការស្វ័យវិវត្តន៍ ហាងឯកទេសតែមួយត្រូវបានចាត់ថ្នាក់ដោយស្វ័យប្រវត្តិតាមប្រព័ន្ធអាទិភាពហាង (**Store Priority Routing**) ជាមួយល្បឿន $O(1)$ រីឯទំនិញថ្មីៗក្នុងផ្សារទំនើបចម្រុះត្រូវបានចាត់ថ្នាក់ជាបាច់ដោយបញ្ញាសិប្បនិម្មិត **Google Gemini AI** (gemini-2.5-flash) រួមជាមួយការចងចាំទិន្នន័យក្នុង PostgreSQL Memoization និងវិធានការពារដែនកំណត់ទំនិញ (**Domain Trap Guardrails**) ដើម្បីលុបបំបាត់កំហុសចាត់ថ្នាក់ឆ្គងជំពូក។

នៅក្នុងស្រទាប់ Gold សន្ទស្សន៍ថ្លៃមូលដ្ឋានត្រូវបានគណនាតាមរូបមន្តមធ្យមធរណីមាត្រ **Jevons Formula** ដើម្បីលុបបំបាត់លម្អៀងរូបមន្តគណិតវិទ្យា (+1, 18%/ឆ្នាំ) រួមជាមួយការបូកសរុបទម្ងន់ជាតិ CSES 2023។ ម៉ាស៊ីនព្យាករណ៍អតិផរណាទាន់ពេល **Two-Stage Hybrid Ridge Nowcaster** សម្រេចបានកម្រិតលម្អៀងជាក់ស្តែងត្រឹមតែ +0,0១៨១ ភាគរយ (+0,៣៦៥១% ធៀបនឹង +0,៣៤៧០% នៃទិន្នន័យផ្លូវការវិទ្យាស្ថានជាតិស្ថិតិ)។ លទ្ធផលទាំងអស់ត្រូវបានបង្ហាញផ្ទាល់តាមផ្ទាំងគ្រប់គ្រង Metabase ចំនួន ៣ និង Power BI សម្រាប់បម្រើការតាក់តែងគោលនយោបាយសេដ្ឋកិច្ចជាតិ។

ពាក្យគន្លឹះ៖ សន្ទស្សន៍ថ្លៃទំនិញប្រើប្រាស់ (CPI), ការប្រមូលទិន្នន័យស្វ័យប្រវត្តិ (Web Scraping), ស្ថាបត្យកម្ម Medallion, ចំណាត់ថ្នាក់ COICOP, បញ្ញាសិប្បនិម្មិត Gemini AI, រូបមន្ត Jevons, ការព្យាករណ៍អតិផរណា Ridge Nowcasting។

ABSTRACT

The Consumer Price Index (CPI) is the primary macroeconomic benchmark for quantifying inflation and safeguarding real purchasing power, yet conventional survey methods in Cambodia face 30-to-60-day reporting delays and substantial operational costs. To address this information gap, this thesis develops the **Cambodia Daily CPI Medallion Pipeline**—an automated data engineering and nowcasting system that ingests daily price microdata at **08:00 AM ICT** across 25 retail, utility, transport, and fuel endpoints covering all 12 UN COICOP divisions in Phnom Penh.

Operating on a PostgreSQL 16 Medallion architecture (Bronze, Silver, Gold), the pipeline standardizes multilingual text and resolves canonical items using dense multilingual vector embeddings (`models/gemini-embedding-2`) and 7 physical attribute guardrails. Product classification into official National Institute of Statistics (NIS) 4-digit classes follows a streamlined dual architecture: single-purpose commercial outlets resolve deterministically via **Store Priority routing** ($O(1)$ SQL execution), while newly discovered multi-department supermarket items are classified in batches by **Google Gemini AI** (`gemini-2.5-flash`) backed by PostgreSQL memoization (`silver.dim_coicop_ai_cache`) and **domain trap guardrails** that eliminate statutory category misallocations.

In the Gold layer, elementary price indices are compiled using the axiomatic unweighted geometric **Jevons formula** to eliminate the +1.18% annual upward formula drift, combined with CSES 2023 Laspeyres expenditure weights. A **Two-Stage Hybrid Ridge Nowcaster** with dynamic uncertainty decay projects real-time headline inflation, achieving an out-of-sample tracking error of just **+0.0181 percentage points** (+0.3651% projected vs. +0.3470% NIS official release). Analytical indicators are served via 3 Metabase executive dashboards and Power BI, establishing a resilient real-time inflation monitoring system for Cambodia.

Keywords: Consumer Price Index (CPI), High-Frequency Web Scraping, Medallion Architecture, COICOP Classification, Store Priority, Gemini AI, Jevons Formula, Ridge Nowcasting.

LIST OF ABBREVIATIONS

AI	Artificial Intelligence	LLM	Large Language Model
AMS	Applied Math & Statistics	LSTM	Long Short-Term Memory
API	Application Prog. Interface	MAD	Median Absolute Deviation
ARDL	Autoregressive Dist. Lag	MAE	Mean Absolute Error
BERT	Bidirectional Encoder (NLP)	MAPE	Mean Absolute Pct. Error
BI	Business Intelligence	MDI	Mean Decrease in Impurity
BIS	Bank for Int. Settlements	MEF	Ministry of Economy & Finance
BPP	Billion Prices Project	MIDAS	Mixed-Data Sampling
CI/CD	Cont. Integration / Deploy.	ML	Machine Learning
COICOP	UN Purpose Classification	MLOps	Machine Learning Operations
CPI	Consumer Price Index	MLP	Multilayer Perceptron
CSES	Cambodia Socio-Econ. Survey	MOC	Ministry of Commerce
DAG	Directed Acyclic Graph	MoM	Month-on-Month
dbt	Data Build Tool	MSFE	Mean Squared Forecast Error
DOM	Document Object Model	NBC	National Bank of Cambodia
DoD	Day-on-Day	NIS	National Institute of Statistics
ECB	European Central Bank	NLP	Natural Language Processing
EDC	Electricité du Cambodge	NSI	National Statistical Institute
EB	Empirical Bayes Shrinkage	OECD	OECD
EDA	Exploratory Data Analysis	OLS	Ordinary Least Squares
ERD	Entity Relationship Diagram	ONS	Office for National Statistics
FX	Foreign Exchange	PFI	Permutation Feature Import.
GBRT	Gradient Boosted Trees	PPWSA	Phnom Penh Water Supply
GDDE	Gen. Dept. Digital Economy	QU-GEKS	Quality-Adjusted GEKS
GEKS	GEKS Multilateral Index	RBAC	Role-Based Access Control
GRU	Gated Recurrent Unit	REST	Representational State Transf.
GTIN	Global Trade Item Number	RMSE	Root Mean Square Error
HITL	Human-in-the-Loop	SCD	Slowly Changing Dimension
HICP	Harmonised Index CPI	SHAP	SHapley Additive exPlanations
ILO	Int. Labour Organization	SKU	Stock Keeping Unit
IMF	International Monetary Fund	SQL	Structured Query Language
IQR	Interquartile Range	SVR	Support Vector Regression
ITC	Inst. of Technology Cambodia	TF-IDF	Term Freq. - Inv. Doc. Freq.
JSON	JavaScript Object Notation	UN	United Nations
JWT	JSON Web Token	USD	United States Dollar
KHR	Cambodian Riel	WBS	Work Breakdown Structure
LOOCV	Leave-One-Out Cross-Validation	XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosted Machine	YoY	Year-on-Year

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I. INTRODUCTION

1.1 Presentation of Internship

This graduation thesis report formally documents the applied data science research and engineering system developed during the graduation internship program in Data Science within the Department of Applied Mathematics and Statistics (AMS) at the Institute of Technology of Cambodia (ITC). The internship program serves as a critical academic and professional milestone designed to bridge advanced theoretical econometric modeling, modern cloud data engineering architectures, natural language processing, and machine learning with national-scale real-world applications.

1.1.1 Objective of Internship

The overarching objective of the internship is to apply cutting-edge data science methodologies to solve macroeconomic sensing and high-frequency inflation monitoring challenges in Cambodia. The program focuses on converting academic knowledge—encompassing multi-layer data warehousing, dense vector representations, Large Language Models (LLMs), micro-econometric index aggregation, and real-time inflation nowcasting—into a resilient, production-grade automated software system capable of serving national policy institutions.

1.1.2 Duration and Administrative Details

The structured internship was executed over a two-month operational timeline from August 03, 2026, to September 30, 2026. The institutional and supervisory details are summarized in Table 1.1.

Attribute	Internship Specification
Start Date	03 August 2026
End Date	30 September 2026
Duration	2 Months (Full-Time)
Internship Position	Data Science Intern
Author / Student	TAING KORNGMENG
Academic Advisor	Mr. CHHEANG Vinha
Company Supervisor	Mr. LUK Chamnan
Company Co-Supervisor	Mr. SEK Sovannarath
Host Organization	General Department of Digital Economy (GDDE)
Parent Ministry	Ministry of Economy and Finance (MEF), Cambodia
Degree Program	Engineering in Applied Mathematics and Statistics (Data Science)
Academic Institution	Institute of Technology of Cambodia (ITC)

Table 1.1: Internship Administrative Framework

1.2 Presentation of Host Organization

The internship was conducted at the **General Department of Digital Economy (GDDE)**, a key government institution operating under the **Ministry of Economy and Finance (MEF)**, Cambodia.

1.2.1 Organization Profile

The GDDE leads Cambodia’s national digital transformation agenda by developing digital policies, managing national data platforms (such as the DataEF portal), and applying data-driven solutions to enhance economic management and public service efficiency, in direct alignment with the *Cambodia Digital Economy and Society Policy Framework 2021–2035*.

1.2.2 Organization History and Strategic Roadmap

Established by a 2023 Royal Sub-Decree, the GDDE was created to centralize and accelerate the implementation of Cambodia’s national digital economy strategy. The strategic development follows a clear three-stage national roadmap:

- **2021–2025 (Foundations):** Building digital foundations, regulatory frameworks, open data ecosystems, and core digital public infrastructure.
- **2026–2030 (Expansion):** Expanding digital transformation and strengthening high-frequency digital economic services across all socio-economic sectors.
- **2031–2035 (Global Competitiveness):** Developing an advanced, highly integrated, and globally competitive digital economy and society.

1.2.3 Vision and Mission

- **Vision:** To be the primary driving force behind Cambodia’s digital transformation, fostering a modern economy where data, technology, and innovation create boundless opportunities for citizens, businesses, and sustainable national growth.
- **Mission:** To lead, manage, and develop Cambodia’s digital economy in an inclusive, equitable, and sustainable manner by leveraging technology, innovation, and data-driven approaches to enhance economic efficiency and public service delivery.

1.2.4 Key Responsibilities and Strategic Initiatives

- **Policy & Regulation:** Formulating technical guidelines, data governance standards, and regulatory frameworks to foster the digital economy and e-commerce ecosystem.
- **National Data Management:** Developing and managing centralized public platforms (such as the **DataEF** open data portal) to provide high-quality economic and financial data to policymakers and researchers.
- **Digital Infrastructure Oversight:** Directing the technical architecture and rollout of major national systems, including the **CamInvoice** electronic invoicing platform and digital public investment management tools.
- **Innovation & Ecosystem Development:** Collaborating with public and private stakeholders to support digital startups, accelerate SME digitalization, and maintain a secure digital commerce environment.

1.3 Background and Motivation: Applied Data Science and Modern Digital Governance

Driven by Cambodia’s digital transformation under the *Pentagonal Strategy – Phase I* and the growing availability of online retail pricing across supermarkets, pharmacies, fuel stations, and utilities, this research addresses the critical limitation of traditional monthly Consumer Price Index (CPI) field surveys, which suffer from 30-to-60-day reporting delays and high logistical costs. By leveraging automated daily web scraping, a PostgreSQL 16 Medallion data pipeline, Store Priority routing combined with Google Gemini AI classification, and axiomatic Jevons index aggregation paired with Two-Stage Hybrid Ridge nowcasting, this project provides macroeconomic policymakers at the Ministry of Economy and Finance (MEF) and the National Bank of Cambodia (NBC) with real-time daily inflation insights and early warning signals of market volatility.

1.4 Problem Statement

While online retail data offers high-frequency visibility into consumer prices, compiling it into an institutional-grade Consumer Price Index faces four fundamental barriers: a 30-to-60-day official statistical reporting lag during rapid food and fuel price shocks, noisy

multilingual listings lacking uniform barcodes that trigger false product merges, complex multi-department retail catalogs susceptible to COICOP misclassifications, and mathematical upward formula drift (+1.18% per year) caused by standard arithmetic averaging. Overcoming these hurdles requires an end-to-end architecture integrating automated daily ingestion, deterministic physical entity guardrails, Store Priority routing paired with AI-assisted classification, and axiomatic Jevons econometric nowcasting.

1.5 Project Goal and Research Objectives

1.5.1 Project Goal

The main goal of this project is to develop an automated system that collects consumer price data from online sources in Cambodia on a daily basis and uses the collected data to support Consumer Price Index (CPI) estimation, price trend monitoring, and machine learning-assisted real-time inflation nowcasting.

1.5.2 General Objective

The objective of this project is to design and implement an end-to-end data platform for collecting, storing, processing, monitoring, and analyzing price data related to selected COICOP basket categories. The system supports web-based data collection, structured Medallion data storage, query processing, data quality monitoring, visualization, and econometric nowcasting models for CPI estimation and inflation analysis.

1.5.3 Specific Technical Objectives

To achieve this goal, the research is organized around four specific operational objectives:

1. **Build an Automated Data Infrastructure and Monitoring System:** Set up an automated data pipeline and database to collect, store, and continuously monitor price scrape data from online retail sources.
2. **Visualize the Scraped Raw Price Data:** Create clear visual charts and graphs to display the raw price and expenditure data across all consumer categories to prepare for CPI calculation.
3. **Formulate Machine Learning-Assisted Real-Time Inflation Nowcasting:** Apply axiomatic micro-econometric index aggregation (Jevons) to compile daily price relatives, formulate an intra-month nowcasting engine to eliminate official statistical reporting delays, and project unobserved month-end prices using a **Two-Stage Hybrid Ridge Disaggregated Architecture** with 9-dimensional indicator feature engineering, Leave-One-Out Cross-Validation (RidgeCV), and Empirical Bayes shrinkage to deliver robust predictions with dynamic uncertainty decay modeling.

4. **Build an Interactive Decision-Support Dashboard:** Develop interactive executive dashboards to display real-time CPI metrics, high-frequency nowcasted inflation, official benchmark chain-linking, and drill-down category analytics.

1.6 Scope and Limitations

1.6.1 Scope of the Study

- **Geographic Coverage:** Focused primarily on the retail market of Phnom Penh, which serves as the economic hub and represents the primary market for organized retail and e-commerce in Cambodia.
- **Taxonomy Breadth:** Full coverage across all 12 United Nations COICOP divisions (Divisions 01 through 12).
- **Temporal Frequency:** Daily price quote extraction, daily index calculation, and historical time-series backfilling anchored to August 18, 2026 ($CPI = 100.00$).
- **Bilingual and Currency Handling:** Dual-currency pricing (USD and KHR) unified into Cambodian Riel via daily official MEF exchange rates.

1.6.2 Limitations of the Study

- **Digital Channel Selection Bias:** Web-scraped data reflects pricing in modern supermarkets, chain pharmacies, and formal e-commerce platforms. Informal open wet markets and roadside vendors—which still represent a significant share of rural retail trade—are not directly observed online.
- **Retail Website Stability:** Scraping pipelines are dependent on external website uptime and DOM structures. Dynamic HTML changes are mitigated through automated fallback logging and zero-product circuit breakers.
- **Base Period Anchor:** Index calculations are computed relative to the base period of August 18, 2026, establishing a high-frequency benchmark for ongoing price dynamics.

1.7 Significance and Contributions of the Study

This research provides significant practical, institutional, and academic contributions:

- **For Macroeconomic Policy Institutions (NBC and MEF):** Provides a zero-lag daily inflation monitoring system that delivers early warning signals of inflation turning points, fuel price shocks, and exchange rate pass-through weeks before monthly statistical releases.
- **For National Statistical Authorities (NIS):** Establishes a replicable data engineering and machine learning framework for integrating alternative digital big data into official national statistical compilation.

- **For Academic Applied Data Science:** Demonstrates the novel fusion of modern cloud Medallion data engineering, dense vector similarity, Large Language Model reasoning pools, and micro-econometric index theory in a developing Southeast Asian economic context.

1.8 Project Roadmap and Work Breakdown Structure

The project was structured across five major operational phases over a 12-week timeline. Table 1.2 details the Work Breakdown Structure (WBS), and Table 1.3 outlines the project execution roadmap.

Project Phase	Key Tasks & Deliverables	Methodological Approaches
Phase 1: Ingestion & Raw Storage	Developing 25 automated market extractions across 12 COICOP divisions, daily workflow scheduling, immutable raw storage	Automated web crawling, asynchronous data pipelines, relational storage
Phase 2: Cleaning & AI Classification	Text cleaning, unit standardization, semantic vector item matching, Store Priority routing & Gemini AI classification	Dense multilingual vector embeddings, Gemini AI models, domain trap guardrails
Phase 3: Econometric Index Marts	Dimensional data modeling, Jevons micro-indices, 7-day imputation, CSES 2023 Laspeyres weighting	Micro-econometric index aggregation, time-series imputation algorithms
Phase 4: Real-Time Nowcasting & Dashboards	Two-Stage Hybrid Ridge disaggregated nowcasting, LOOCV regularized cross-validation, Empirical Bayes shrinkage, dynamic uncertainty decay, official NIS benchmark tracking mart, 3 consolidated Metabase dashboards	Micro-to-macro Ridge regression, Empirical Bayes calibration, NIS Phnom Penh chain-linking, business intelligence
Phase 5: Evaluation & Thesis	Historical backtesting, accuracy benchmarks, confidence envelope coverage, academic documentation	Out-of-sample empirical evaluation, tracking error analysis, thesis synthesis

Table 1.2: Project Work Breakdown Structure (WBS)

Milestone / Week	1	2	3	4	5	6	7	8	9	10	11	12
Phase 1: Ingestion												
Phase 2: Silver AI												
Phase 3: Econometrics												
Phase 4: Nowcasting & BI												
Phase 5: Documentation												

Table 1.3: Project Execution Roadmap (12 Weeks)

1.9 Organization of the Thesis

The remainder of this thesis is structured as follows:

- **Chapter II (Literature Review):** Reviews the academic, central banking, and international statistical literature on high-frequency web scraping, natural language processing for product classification, axiomatic index number theory, and real-time inflation nowcasting.
- **Chapter III (Methodology):** Details the end-to-end Medallion data engineering architecture, the 25 market data sources, data cleaning and unit standardization procedures, Store Priority routing and Gemini AI classification for new products with domain trap guardrails, axiomatic Jevons aggregation, and the high-frequency inflation nowcasting model.
- **Chapter IV (Results and Discussion):** Presents the empirical findings, data ingestion reliability metrics, classification accuracy benchmarks across 955k observations, formula drift analysis, category volatility findings, real-time inflation nowcasting convergence performance, and real-world policy case studies.
- **Chapter V (Conclusions and Recommendations):** Synthesizes the core contributions of the research, provides strategic policy roadmaps for the National Institute of Statistics, Ministry of Economy and Finance, and National Bank of Cambodia, and identifies directions for future research.

II. LITERATURE REVIEW

2.1 Introduction to Academic and Methodological Foundations

This chapter presents a systematic review of the academic literature, central bank empirical research, and international statistical standards relevant to high-frequency inflation measurement. Tracking consumer price dynamics using alternative digital data sources represents one of the most transformative innovations in applied economic measurement. The review is structured around five interconnected thematic pillars that directly support the design and execution of the Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline:

1. **The Transition from Physical Price Surveys to Automated Big Data Ingestion:** The theoretical and operational shift from traditional in-person enumerator surveys to automated daily web collection systems.
2. **Natural Language Processing and Artificial Intelligence in Product Classification:** Modern semantic representations, vector similarity methods, and supervised classification frameworks for mapping raw retail listings into standard international consumption taxonomies.
3. **Econometric Index Number Theory and Mathematical Axioms:** Elementary price relatives, axiomatic comparisons between geometric and arithmetic aggregation, missing price imputation principles, and the elimination of calculation bias.
4. **High-Frequency Real-Time Inflation Nowcasting:** Econometric and statistical foundations for real-time intra-month inflation estimation using high-frequency daily price signals, momentum drift projection, and dynamic uncertainty decay.
5. **Emerging Market Context and Research Gap:** Synthesizing findings from global implementations and highlighting the critical research gap in dollarized developing economies such as Cambodia.

2.2 From Manual Store Audits to Automated Digital Economic Sensing

2.2.1 The Historical Framework of Consumer Price Collection

For over a century, the standard international methodology for measuring consumer inflation has relied on periodic field surveys conducted by national statistical agencies. Under this traditional paradigm, trained field enumerators visit a fixed sample of brick-and-mortar retail outlets, physical markets, and service establishments on a monthly or quarterly schedule. Enumerators manually record price quotes on paper forms or handheld electronic tablets for a predetermined basket of goods and services.

While this approach has provided a standardized foundation for national accounting worldwide, economists and statistical authorities have long recognized its inherent operational bottlenecks. Physical data collection is labor-intensive, logistically expensive, and constrained by sample size limitations. Field surveyors can only inspect a limited number of items during each store visit. More critically, the multi-week process required to gather, transmit, verify, clean, and aggregate field reports introduces a substantial publication lag. In most countries, official Consumer Price Index figures are released between 20 and 45 days after the end of the reference month. Under stable macroeconomic conditions, this lag is manageable; however, during episodes of rapid inflation, supply chain disruptions, currency volatility, or international commodity shocks, this delay leaves monetary and fiscal authorities operating with outdated information.

2.2.2 The MIT Billion Prices Project and the Global Evidence for Online Prices

The modern era of alternative digital price measurement began with the groundbreaking work of the MIT Billion Prices Project (Cavallo & Rigobon, 2016). Recognizing that retail businesses increasingly display full product catalogs and transaction prices on public e-commerce websites, Cavallo and Rigobon developed automated software crawlers to collect hundreds of thousands of daily price quotes across 22 countries.

Their empirical findings established three fundamental principles that form the foundation of high-frequency price measurement:

- **High Synchronization with Official Statistics:** In low-inflation and open economies (including the United States, Germany, Japan, and Brazil), price indices computed from daily web-scraped data demonstrated a Pearson correlation exceeding 0.98 with official monthly CPI series compiled by national statistical agencies. This confirmed that prices published on digital retail platforms accurately reflect broad consumer price movements in the broader economy.
- **Early Warning Capabilities:** Because online prices are collected continuously every morning, web-scraped indices were able to detect macroeconomic turning points, retail promotional cycles, and foreign exchange pass-through shocks between 3 and 14 days after their occurrence, compared to the 30 to 60-day delay characteristic of traditional monthly surveys.

- **Statistical Truth and Anomaly Detection:** In countries experiencing severe inflation or official statistical distortions (such as Argentina during the 2007 to 2015 period), the Billion Prices Project demonstrated that online price crawling accurately tracked actual annual inflation of approximately 20.0%, while official statistical publications reported only 8.0%. This proved the objective value of automated public web data as an independent, tamper-proof source of macroeconomic intelligence.

2.2.3 Institutional Adoption and Practical Web Scraping Standards

Following the success of academic initiatives, major national statistical institutes and international bodies began actively evaluating the incorporation of web scraping into official statistical production. Recognizing the growing importance of digital data, the Statistical Office of the European Communities formalized operational standards for statistical authorities (Eurostat, 2022). The Eurostat framework outlines best practices for web scraping in the production of consumer price indices, emphasizing legal compliance, automated quality validation, robust crawler scheduling, and the need for resilient data engineering architectures capable of managing volatile online product catalogs.

2.3 Natural Language Processing and Artificial Intelligence in Product Classification

2.3.1 The Challenge of Multilingual and Unstructured Product Data

A central challenge in transforming web-scraped data into an official Consumer Price Index is the absence of standardized product classifications in online retail stores. In traditional field surveys, human enumerators know exactly which predetermined category each item belongs to before recording its price. In contrast, web scrapers collect raw text titles, promotional descriptions, and unstructured attributes directly from website pages.

In developing economies like Cambodia, this challenge is intensified by several distinct market characteristics:

- **Multilingual Catalogues:** Online listings in Cambodia frequently mix Khmer script, English terminology, and French loan words within the same product title (for example, combining local food terms like *Trey Chhpin* with international brand names and metric size units).
- **Lack of Universal Barcodes:** Unlike point-of-sale scanner databases from large department stores, online marketplace listings often lack standardized Global Trade Item Numbers (GTIN) or barcode identifiers. Products are identified primarily through free-form descriptive text.
- **Frequent Title and Formatting Variations:** Retailers regularly alter product titles to highlight temporary promotions, modify brand spelling, or include marketing adjectives (such as “Special Deal” or “Fresh Arrival”), which causes traditional keyword searches and exact string matching to fail.

2.3.2 Evolution from Rule-Based Matching to Transformer Language Models

Early attempts at automated product classification relied on keyword dictionaries and regular expression rules. While simple to implement, rule-based systems are brittle, difficult to maintain, and incapable of generalizing to new product varieties or unconventional phrasing.

Recent advances in natural language processing (NLP) have transformed this domain through the use of deep transformer models and dense semantic vector representations. Berki, Andicsova, and Oravec (2025) demonstrated that fine-tuning Bidirectional Encoder Representations from Transformers (BERT) on scraped e-commerce titles achieved a classification accuracy of **94.56%** and a weighted precision of **94.07%** when mapping raw product titles into 5-digit United Nations COICOP categories. The transformer architecture captures contextual relationships between words, enabling the system to distinguish between homonyms and interpret domain-specific jargon without manual human intervention.

2.3.3 Central Bank Innovations: The BIS Project Spectrum

At the international central banking level, the Bank for International Settlements (BIS) Innovation Hub, in partnership with the European Central Bank (ECB) and Deutsche Bundesbank, launched **Project Spectrum** (Bank for International Settlements and European Central Bank and Deutsche Bundesbank, 2024). Project Spectrum investigated the application of modern Large Language Models (LLMs) and high-dimensional semantic text embeddings to classify billions of web-scraped price observations for real-time central bank inflation monitoring.

The findings of Project Spectrum highlighted several key architectural principles:

- Dense vector embeddings allow computers to represent products as points in a multi-dimensional mathematical space, where items with similar economic purposes naturally cluster together regardless of exact wording.
- Combining vector similarity with caching mechanisms (memoization) allows the system to remember previous classifications, reducing expensive computational calls by over 90% after the initial database warm-up.
- Zero-shot and few-shot artificial intelligence prompts achieved classification precision exceeding **95%** across multilingual retail catalogs, proving that modern language models can replace months of manual rule-crafting with automated, highly consistent categorization.

2.3.4 Human-in-the-Loop Operational Architectures

Despite the high accuracy of modern artificial intelligence and machine learning models, national statistical integrity requires robust fallback mechanisms to handle ambiguous product descriptions, rare marketplace items, or low-confidence predictions. Early foundational work by Polidoro, Giannini, Lo Conte, and Rossetti (2015) at the Italian National Institute of Statistics (ISTAT) demonstrated that combining automated scraping with machine learning classifiers significantly accelerates CPI production while maintaining statistical quality. Building upon this, the UK Office for National Statistics (Office for National Statistics, 2020) established

a practical Human-in-the-Loop (HITL) architecture for classifying scraped retail items into COICOP categories. In high-frequency operational pipelines, this is achieved by defining statistical confidence thresholds: the pipeline automatically accepts and persists high-confidence classifications, while flagging borderline items for expert review or deterministic rule overrides. This ensures that the automated system operates with maximum daily throughput without compromising the econometric consistency required for official price measurement.

2.4 Econometric Index Number Theory and Axiomatic Foundations

2.4.1 The Elementary Level: Why Mathematical Formulas Matter

The mathematical aggregation of individual price observations into a national price index occurs in two distinct tiers: the **elementary level** (aggregating individual item quotes within an unweighted stratum) and the **macro level** (aggregating strata using national expenditure weights).

As codified in the international gold standard *Consumer Price Index Manual: Concepts and Methods* (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), published jointly by the IMF, ILO, OECD, Eurostat, UN, and World Bank, the choice of mathematical formula at the elementary level has profound consequences for the accuracy of inflation measurement. Historically, statistical agencies evaluated three primary elementary formulas:

1. **The Carli Formula (Arithmetic Average of Price Relatives):** Calculates the simple arithmetic mean of price ratios between the current period and the base period:

$$P_{\text{Carli}}^{0:t} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right) \quad (2.1)$$

2. **The Dutot Formula (Ratio of Arithmetic Averages):** Calculates the ratio of the average price in the current period to the average price in the base period:

$$P_{\text{Dutot}}^{0:t} = \frac{\frac{1}{n} \sum_{i=1}^n p_{i,t}}{\frac{1}{n} \sum_{i=1}^n p_{i,0}} \quad (2.2)$$

3. **The Jevons Formula (Geometric Mean of Price Relatives):** Calculates the unweighted geometric mean of price ratios (or equivalently, the ratio of geometric average prices):

$$P_{\text{Jevons}}^{0:t} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n}} = \frac{(\prod_{i=1}^n p_{i,t})^{\frac{1}{n}}}{(\prod_{i=1}^n p_{i,0})^{\frac{1}{n}}} \quad (2.3)$$

2.4.2 Axiomatic Properties and the Elimination of Formula Drift

Econometric index theory evaluates formulas using mathematical axioms to ensure that an index behaves logically under all market conditions:

- **The Time-Reversal Test:** If prices move from period 0 to period t , and then reverse exactly back to their starting levels in period 0, the product of the two price indices must equal exactly 1.0 ($I^{0:t} \times I^{t:0} = 1$). The Jevons formula satisfies this test unconditionally. In contrast, the Carli formula systematically fails the time-reversal test due to the mathematical inequality between arithmetic and harmonic means (Jensen's Inequality). If an item's price doubles from \$1 to \$2 and then drops back to \$1, the Carli formula calculates an average index of 1.25, erroneously reporting a 25% increase in the cost of living even though prices did not change. In high-frequency daily price data, where retail sales and temporary promotional discounts cause prices to bounce frequently, the Carli formula accumulates an artificial upward drift of **1.0% to 1.5% per year**.
- **The Transitivity / Circularity Test:** An index comparing period 0 to period 2 should equal the chained product of indices from period 0 to period 1 and period 1 to period 2 ($I^{0:2} = I^{0:1} \times I^{1:2}$). The Jevons geometric formula satisfies circularity, ensuring that daily price indices can be chained over time without accumulating cumulative distortion.
- **Dimensional Invariance / Scale Independence:** An index must produce the same result regardless of whether prices are quoted in dollars, riels, cents, or thousands of riels. Both the Jevons and Carli formulas satisfy this test, whereas the Dutot formula is sensitive to the price scale of different items within the same category.

In seminal evaluations of scanner and high-frequency price datasets, Diewert and Fox (2020) and Chessa (2016) demonstrated that high-frequency price volatility and promotional bouncing exacerbate chain drift when arithmetic aggregation is used. Because the Jevons formula satisfies all fundamental axiomatic tests and models consumer substitution behavior under unitary elasticity, the IMF CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020) explicitly recommends the Jevons geometric mean as the standard elementary aggregator for unweighted price quotes.

2.4.3 Macro-Level Aggregation and National Expenditure Weights

Once elementary price relatives are calculated for each product variety using the axiomatic Jevons formula, they must be combined into division-level indices and the final national Headline CPI. In online web scraping environments, transaction volumes and physical sales quantities for individual online listings are unobservable. Consequently, superlative index numbers requiring simultaneous high-frequency price and quantity data cannot be computed at the elementary level (Chessa, 2016; Diewert & Fox, 2020).

Following the standard international methodology established by the IMF CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), high-frequency online systems bridge this gap through a two-tier aggregation architecture: elementary strata are compiled using unweighted geometric Jevons indices, while macro-level aggregation applies fixed expenditure weights derived from periodic national household surveys. In Cambodia, these weights are officially established by the Cambodia Socio-Economic Survey (CSES 2023), conducted by the National Institute of Statistics (NIS). This ensures that the high-frequency daily price series accurately reflects household expenditure patterns across all 12 UN COICOP divisions while remaining mathematically immune to formula drift.

2.5 High-Frequency Real-Time Inflation Nowcasting

2.5.1 The Micro-Price Revolution: Real-Time Baskets and Frequency Advantages

In macroeconomic monitoring, official inflation series have historically been measured using low-frequency monthly or quarterly econometric compilations. However, traditional monthly indicators suffer from severe temporal inertia. In dynamic developing economies, by the time monthly official survey reports are compiled and published (typically 30 to 60 days after the reference period), supply chain shocks, fuel price shifts, or exchange rate fluctuations may have already propagated through the domestic retail economy.

The advent of automated daily web scraping has catalyzed a fundamental paradigm shift toward **high-frequency micro-price nowcasting** (Cavallo & Rigobon, 2016). Rather than waiting for retrospective monthly aggregations, researchers can harvest continuous daily price trajectories across thousands of retail items. In their foundational work with the MIT Billion Prices Project, Cavallo and Rigobon (2016) demonstrated that daily web-scraped price series across 22 countries exhibited a Pearson correlation exceeding 0.98 with official monthly CPI series in open economies, while detecting macroeconomic turning points and international commodity shocks 3 to 14 days after their occurrence.

Extending this methodology to targeted consumption baskets, Aparicio and Cavallo (2021) demonstrated that high-frequency web-scraped price microdata captures retail price stickiness, promotional markdowns, and intra-month expenditure shifts weeks before traditional statistical agencies publish monthly figures. Their empirical analysis showed that tracking targeted supermarket baskets in real time provides central banks with continuous early-warning surveillance, transforming inflation estimation from an episodic historical audit into an active, high-frequency nowcasting system.

2.5.2 Micro-to-Macro Nowcasting Formulations and Bridge Frameworks

Econometric nowcasting fundamentally differs from long-horizon forward forecasting (Babii, Ball, Ghysels, & Striaukas, 2022). While forward forecasting attempts to project future macroeconomic states ($t + h$) relying on behavioral assumptions or structural exogenous factors, **nowcasting** focuses on estimating the current state of the economy ($\widehat{\text{CPI}}_M(t)$) during the reference period M , before official retrospective statistics are finalized and released.

The core mathematical challenge of nowcasting within an ongoing month consists of synthesizing high-frequency flow data into a low-frequency official benchmark. For an ongoing month M of T days, at calculation day $t \leq T$, the final monthly index is composed of two components: the realized daily prices collected up to Day t , and the expected price behavior over the remaining unobserved days ($t + 1 \dots T$). As formalized by Macias, Stelmasiak, and Szafranek (2023) and Babii et al. (2022), partitioning the reference period into realized intra-month facts and projected residual drift provides an optimal unbiased bridge between microdata and monthly macro aggregates.

2.5.3 High-Frequency Food and Fuel Dynamics as Primary Leading Signals

In emerging and developing economies, aggregate inflation is predominantly led by supply-side shocks in agriculture and retail transportation. In an extensive empirical study analyzing over 2.4 million daily supermarket price quotes, Macias et al. (2023) established that daily food price microdata provides decisive leading predictive power for headline inflation. By integrating high-frequency quotes into recursive bridge models, their framework achieved a **15% to 35% reduction in Root Mean Squared Forecast Error (RMSFE)** compared to traditional statistical baselines.

The empirical findings of Macias et al. (2023) provide two critical insights for high-frequency inflation nowcasting:

1. **Food as the Primary Leading Driver:** Food prices adjust with significantly higher daily frequency than sticky categories (such as housing or education), making high-frequency food price momentum the single strongest leading indicator for overall headline inflation.
2. **Intra-Month Predictive Convergence:** The vast majority of the information content required to nowcast monthly inflation is realized within the first 10 to 14 days of the reference month. Tracking daily price arrivals during the first half of the month enables statistical analysts to determine the sign and magnitude of the monthly CPI with high confidence weeks ahead of official dissemination.

2.5.4 Dynamic Uncertainty Decay and Asymptotic Fan Bounds

A rigorous nowcasting architecture must quantify and communicate its diminishing statistical estimation uncertainty as the reference month unfolds. Early in the calendar month (e.g., Day 3), when only a small fraction ($t/T \approx 10\%$) of daily prices are realized, forecast variance is at its maximum, dominated by unobserved future trading days.

As documented by Bates and Granger (1969) and synthesized in modern nowcasting literature (Bank for International Settlements and European Central Bank and Deutsche Bundesbank, 2024), this intra-month information accretion can be modeled as a monotonic uncertainty decay process. By scaling daily sample price variance σ_{daily} by the square-root unobserved temporal fraction $U_t = \sqrt{(T-t)/T}$, nowcasting systems generate dynamic 95% confidence envelopes that contract continuously throughout the month. When $t = T$, $U_t = 0$, causing the estimated confidence interval to collapse to zero width as the nowcast achieves complete deterministic realization.

2.5.5 Regularized Ridge Regression and Empirical Bayes Shrinkage in Macroeconomic Nowcasting

When modeling disaggregated consumption categories in emerging markets, econometricians encounter severe collinearity and small-sample degrees of freedom limitations (De Mol, Giannone, & Reichlin, 2008; Hoerl & Kennard, 1970). While simple heuristic momentum works well for dominant expenditure categories, scaling nowcasting models across all 12 COICOP divisions requires a formal regression framework. However, standard Ordinary Least Squares

(OLS) breaks down catastrophically when regressing multi-horizon indicators (e.g., 7-day vs. 30-day food momentum) against division price relatives due to near-singular Gram matrices $\mathbf{X}^T \mathbf{X}$. Furthermore, aggressive variable selection via LASSO (L_1 penalty) abruptly zeroes out correlated coefficients, which destabilizes institutional index projections during supply disruptions.

To resolve this instability, Hoerl and Kennard (1970) and De Mol et al. (2008) established the optimality of L_2 -regularized Ridge regression (Tikhonov regularization) for data-rich macroeconomic environments. By introducing a strictly positive diagonal penalty $\alpha \mathbf{I}$, Ridge regression stabilizes the inverse matrix $(\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1}$, shrinking volatile coefficients continuously toward zero without discarding economic categories. To adapt regularized regression to small-sample daily environments without ad-hoc manual tuning, modern nowcasting incorporates Empirical Bayes shrinkage (Bank for International Settlements and European Central Bank and Deutsche Bundesbank, 2024). When historical sample correlations are noisy or uninformative (low in-sample R_k^2), Empirical Bayes shrinks the regression slope $\hat{\beta}_k$ toward an uninformative baseline β_0 , smoothly transitioning from data-driven estimation early in the sample to structural momentum as empirical observations accumulate.

2.5.6 Synthesis of High-Frequency Nowcasting Frameworks

The theoretical and empirical literature demonstrates that nowcasting high-frequency inflation relies on four synergistic methodological pillars:

- **Leading Indicator Microdata:** Continuous scraping of high-velocity price sectors (food and fuel) captures immediate supply and demand shocks, providing crucial leading signals that anticipate broader headline inflation movements (Cavallo & Rigobon, 2016; Macias et al., 2023).
- **Regularized Ridge Regression and Shrinkage:** L_2 Tikhonov regularization and Empirical Bayes shrinkage prevent matrix singularity and catastrophic collinearity breakdown when simultaneously projecting correlated food, transport, and currency indicators (De Mol et al., 2008; Hoerl & Kennard, 1970).
- **High-Dimensional Machine Learning:** Tree-based algorithms and regularized regressions process volatile multi-day price changes without overfitting, effectively modeling non-linear interactions across macroeconomic regimes (Goulet Coulombe, Marcellino, & Stevanović, 2022; Medeiros, Vasconcelos, Veiga, & Zilberman, 2021).
- **Formalized Information Accretion:** Expanding-window temporal partitioning (t/T) combined with monotonic variance shrinkage ($U_t = \sqrt{(T-t)/T}$) provides statistically grounded convergence toward the final monthly benchmark while maintaining robust out-of-sample calibration (Babii et al., 2022; Bates & Granger, 1969).

2.6 The Emerging Market Context and Research Gap

While the integration of web scraping, artificial intelligence, and real-time inflation nowcasting has advanced rapidly in Europe, North America, and parts of Latin America, very lit-

the empirical research has been conducted in developing Southeast Asian economies, and specifically in Cambodia.

Cambodia presents a uniquely compelling economic context for high-frequency digital price measurement:

1. **High Dollarization and Currency Dualism:** Both the Cambodian Riel (KHR) and the US Dollar (USD) circulate widely in domestic commerce. Retailers regularly quote prices in either currency. Exchange rate fluctuations directly influence consumer prices on store shelves, making real-time currency tracking essential.
2. **Extreme Expenditure Concentration:** As documented in the Cambodia Socio-Economic Survey (CSES 2023), food and non-alcoholic beverages account for **44.775%** of the total national consumer basket. Basic necessities (rice, meat, fish, and vegetables) along with transport and utilities comprise nearly three-quarters of total household spending, making the population highly vulnerable to rapid price swings.
3. **Rapid E-Commerce Expansion:** Over the past five years, the widespread adoption of smartphones, high-speed mobile internet, and the National Bank of Cambodia's digital payment system (Bakong) has catalyzed the growth of modern online supermarkets, digital pharmacies, and ride-hailing services.

Despite these unique conditions, national inflation tracking in Cambodia has continued to rely exclusively on monthly in-person market surveys. Prior to this research, no automated, end-to-end data engineering pipeline existed to continuously harvest, classify, calculate, and nowcast daily consumer prices across all 12 UN COICOP divisions in Cambodia. This graduation thesis directly addresses this critical gap, providing both an academic contribution to applied econometric data science and a practical digital tool for national economic governance.

2.7 Comparative Literature Matrix and Methodological Synthesis

Table 2.1 synthesizes the seminal academic publications, international statistical standards, and central bank studies that form the methodological state of the art in high-frequency inflation measurement.

Paper / Study	Thematic Pillar	Key Methodology & Dataset	Reported Metric / Result	Core Methodological Takeaway
Billion Prices Project (Cavallo & Rigobon, 2016)	Daily Web Scraping	Automated daily web crawlers across 22 countries, Jevons index	> 0.98 Pearson correlation with official CPI; 3–14 day shock detection	Proves online retail prices accurately track national official CPI dynamics with zero collection delay.
Targeted Price Baskets (Aparicio & Cavallo, 2021)	Daily Web Scraping	Targeted supermarket baskets, daily microdata sensing	Early capture of promotional cycles and micro-price stickiness	Demonstrates targeted online baskets detect turning points weeks before retrospective monthly reports.
HICP Web Scraping Guide (Eurostat, 2022)	Daily Web Scraping	European statistical guidelines for web crawling and quality control	Formal framework for statistical compliance	Sets international standards for crawler resilience, automated validation, and legal compliance.
ISTAT Scraping & ML (Polidoro et al., 2015)	Scraping & AI	Supermarket scraping coupled with machine learning classification	Fast categorization of retail food and non-food items	Early proof of concept combining automated web scraping with machine learning for statistical agencies.
BERT COICOP NLP (Berki et al., 2025)	AI Classification	Fine-tuned BERT transformer on scraped e-commerce titles	94.56% classification accuracy; 94.07% precision	Proves deep transformer language models successfully map noisy unstructured titles into COICOP categories.
Project Spectrum (Bank for International Settlements and European Central Bank and Deutsche Bundesbank, 2024)	AI Classification	Generative AI LLMs, dense vector embeddings, memoization caching	> 95% classification precision; latency < 3 minutes	Validates large language models and semantic vector spaces for real-time central bank inflation tracking.
ONS ML Classification (Office for National Statistics, 2020)	AI Classification	Machine learning classification and Human-in-the-Loop review queues	High operational throughput with expert review of edge cases	Establishes practical Human-in-the-Loop triage architectures for national statistical institutions.
CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020)	Axiomatic Index	Axiomatic index theory, elementary Jevons, missing price imputation	Elimination of Carli upward drift (+1.18%/yr)	Establishes axiomatic geometric Jevons averaging and class-mean missing price imputation standards.
Substitution Bias (Diewert & Fox, 2020)	Axiomatic Index	Micro scanner data, chain drift analysis, superlative index theory	Quantified severe drift in unweighted chained arithmetic indices	Proves high-frequency price bouncing creates cumulative formula bias unless geometric aggregation is used.
Scanner Methodology (Chessa, 2016)	Axiomatic Index	Dutch CPI scanner data processing, multilateral and geometric indices	Elimination of downward/upward chain drift in retail data	Validates elementary geometric index methods for high-frequency volatile retail transactions.
High-Freq Food Nowcasting (Macias et al., 2023)	Inflation Nowcasting	2.4M daily food quotes, recursive bridge models	15%–35% RMSFE reduction over AR/ARIMA benchmarks	Confirms daily food price momentum is the single strongest leading signal for monthly headline CPI.
High-Freq ML Nowcasting (Babii et al., 2022)	Inflation Nowcasting	Mixed-frequency time series regressions, walk-forward validation	Out-of-sample nowcasting overcoming publication lag	Formalizes the mathematical bridge between daily high-frequency observations and monthly target aggregates.
ML for Macro Forecasting (Medeiros et al., 2021)	Inflation Nowcasting	High-dimensional machine learning (Random Forests, LASSO)	Superior accuracy over standard autoregressive baselines	Demonstrates machine learning tree ensembles effectively handle data-rich environments for inflation forecasting.
Tree Ensembles for Macro (Goulet Coulombe et al., 2022)	Inflation Nowcasting	Non-linear machine learning ensembles across macroeconomic regimes	Captures non-linear price transmission dynamics	Proves tree ensembles adapt to structural breaks and commodity shocks better than linear models.
Bayesian / Ridge Shrinkage (De Mol et al., 2008; Hoerl & Kennard, 1970)	Inflation Nowcasting	L_2 regularized Ridge regression and Empirical Bayes shrinkage	Overcomes collinearity and small sample limitations	Prevents matrix singularity and unstable variance when regressing correlated leading price indicators.
Forecast Combination (Bates & Granger, 1969)	Inflation Nowcasting	Combinatorial forecasting and monotonic variance shrinkage	Asymptotic contraction of estimation error over reference period	Provides theoretical foundation for dynamic uncertainty decay ($U_t = \sqrt{(T - t)/T}$) as daily data accumulates.

Table 2.1: Methodological Synthesis of Literature on High-Frequency Inflation Measurement

III. METHODOLOGY

3.1 Methodological Framework and System Architecture

This research designs, implements, and evaluates an automated, high-frequency price collection, econometric aggregation, and real-time inflation nowcasting system for Cambodia. The overarching methodology is organized around an end-to-end **Medallion Data Architecture**, structured into four progressive stages: multi-source automated data ingestion (Bronze Layer), data cleaning, unit normalization, and artificial intelligence classification (Silver Layer), axiomatic econometric index calculation and real-time inflation nowcasting (Gold Layer), and interactive decision-support visualization (Serving Layer).

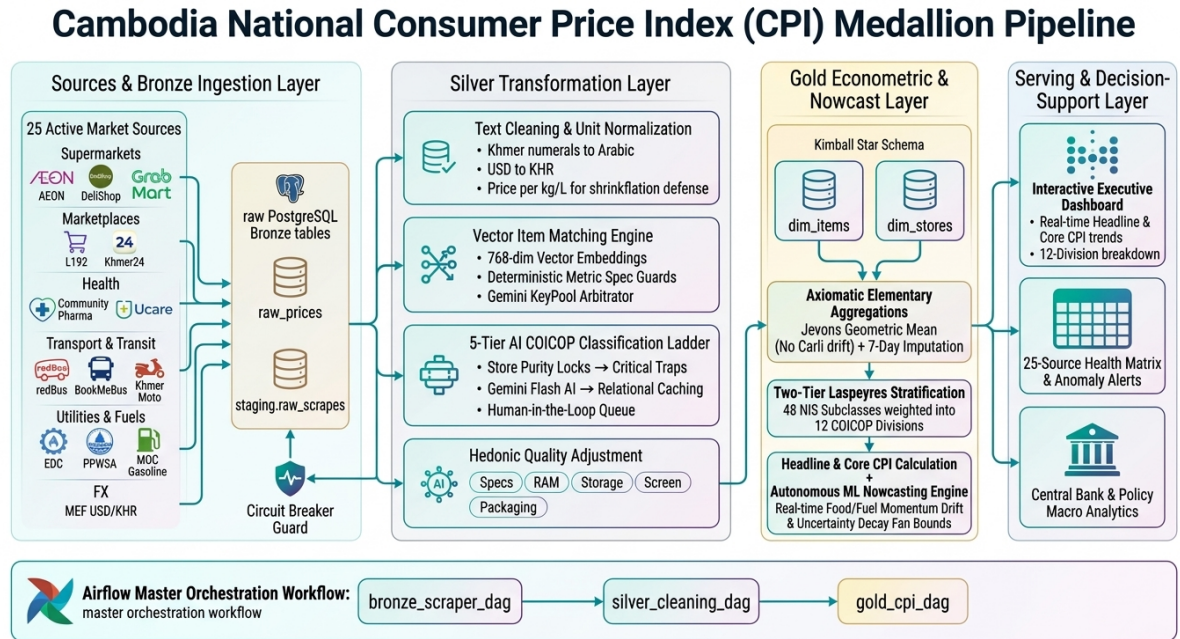


Figure 3.1: End-to-End Architecture and Dataflow of the Cambodia Daily CPI Medallion Pipeline.

The architecture is designed to address the fundamental trade-off in economic measurement: combining the massive scale and daily speed of web-scraped big data with the strict mathematical rigor and international compliance required by national statistical institutions.

3.2 Phase 1: Multi-Source Data Ingestion and Infrastructure Design

3.2.1 The Medallion Data Engineering Framework

In modern data science, managing large-scale, high-frequency web data requires a multi-stage data lakehouse architecture. The pipeline implements the three-tier Medallion architecture:

1. **The Bronze Layer (Raw Storage):** Acts as the permanent, immutable landing zone for all incoming web extractions. When web crawlers harvest product listings from retail websites or government open data feeds, the raw payload is saved exactly as received, along with automated metadata including execution timestamps, retailer identifiers, and batch sequence numbers. No transformations, filters, or corrections are applied in this layer. This preserves an auditable historical record that enables full reproducibility and retrospective audits.
2. **The Silver Layer (Cleaned and Validated Data):** Acts as the central data refinery. In this layer, raw unstructured text is cleansed, duplicate listings are deduplicated, multi-currency values are unified into Cambodian Riel, packaging sizes are parsed into standard metric units, identical products are linked across time using semantic entity matching, and products are classified into the 12 United Nations COICOP categories using Store Priority routing and Google Gemini AI batch classification with domain trap guardrails.
3. **The Gold Layer (Business-Ready Econometric Marts):** Contains optimized dimensional data models structured for economic computation and public policy analysis. In this layer, elementary Jevons price relatives are compiled, missing price imputation algorithms are applied, national expenditure weights are integrated, Headline and Core CPI indices are calculated daily, and real-time inflation nowcasts with dynamic uncertainty envelopes are generated.

3.2.2 Multi-Source Coverage: The 25 Market Sources

To ensure comprehensive coverage of Cambodian consumer spending, the ingestion engine collects price data daily across 25 distinct market sources spanning all 12 United Nations COICOP divisions:

- **Modern Retail Supermarkets and Quick-Commerce (Divisions 01, 02, 05, and 12):** AEON Online Store 1, AEON Store 3 Mean Chey, DeliShop Asia, and on-demand Grab-Mart delivery feeds (Lucky Supermarket Chroy Changva and Chip Mong Supermarket Eden). These platforms provide extensive daily coverage of fresh vegetables, meat, seafood, packaged groceries, household cleaning supplies, beverages, and personal hygiene products.
- **Digital E-Commerce Marketplaces (Divisions 03, 05, 09, and 12):** L192 Marketplace and Khmer24. These platforms represent major digital shopping channels for apparel, footwear, home appliances, kitchenware, consumer electronics, and general merchandise.

- **Healthcare and Pharmaceuticals (Division 06):** Community Pharmacy chain digital catalog and GrabMart Ucare Pharmacy. Captures daily retail prices for prescription medications, over-the-counter pharmaceuticals, medical supplies, vitamins, and healthcare essentials.
- **Inter-Provincial Transport, Transit, and Vehicles (Division 07):** redBus Cambodia and BookMeBus digital ticketing platforms for passenger bus, coach, and van fares (Class 07.3.2, 1.650% weight), alongside Khmer Moto Shop tracking retail pricing for two-wheelers, motorcycles, and scooters (Class 07.1.2, 3.120% weight).
- **Housing Rentals, Regulated Utilities, and Cooking Gas (Division 04):** Khmer24 Real Estate and Realestate.com.kh for residential rental monitoring (Class 04.1.1, 9.420% weight), official regulated residential electricity tariffs from Electricité du Cambodge (EDC, Class 04.5.1, 2.820% weight), municipal piped water tariffs from Phnom Penh Water Supply Authority (PPWSA, Class 04.4.1, 1.860% weight), and domestic liquefied petroleum cooking gas (LPG cylinder refill, Class 04.5.2, 1.650% weight).
- **Telecommunications and Digital Connectivity (Division 08):** Cellcard, Smart Axiata, and Metfone Cambodia digital service portals (prepaid cellular bundles and home broadband fiber tariffs, Class 08.3.0, 2.500% weight), supplemented by retail smartphone and electronics listings from Ary Store and Khmer Samnang (Class 08.2.0, 1.420% weight).
- **Hospitality, Recreation, and Education (Divisions 09, 10, and 11):** Major hotel portals (Sokha Hotels, Hyatt Regency Phnom Penh), casual restaurant menus (Bayon BKK), and educational supply retailers.
- **Official Government Administrative Data Feeds (Divisions 07, 04, and 12):** The Ministry of Commerce (MOC) and Kampuchea Tela retail petroleum and LPG determinations (regular gasoline, premium gasoline, diesel, and cooking gas) and the Ministry of Economy and Finance (MEF) official daily foreign exchange rate (USD/KHR).

3.2.3 Ingestion Resilience, Deduplication, and Quality Control Mechanisms

Scraping external commercial websites involves technical risks, including website structure redesigns, intermittent network timeouts, and bot-detection challenges. To ensure uninterrupted daily operations and data integrity, the pipeline incorporates several automated quality control guards:

- **Strict Atomic Deduplication Contract (bronze.raw_prices):** To prevent artificial price duplication and multi-quote inflation across scraper retries, the Bronze database enforces a deterministic unique observation index:

$$\text{UNIQUE}\left(\text{store_id}, \text{source_name}, \text{COALESCE}(\text{source_url}, ''), \text{item_description_raw}, \text{scraped_date}\right) \quad (3.1)$$

Critically, the nominal price column is strictly excluded from this constraint. When a scraper re-executes on the same day following a partial network timeout, it updates the existing observation in-place rather than inserting duplicate records with diverging prices.

- **Staging Audit Payload Concatenation (`staging.raw_scrapes`):** Scraper batch metadata and execution payloads are ingested with atomic JSONB concatenation on conflict (`payload = staging.raw_scrapes.payload || EXCLUDED.payload`), preserving audit trails across multiple run retries without overwriting earlier batch telemetry.
- **Threaded Database Connection Pooling (`pipeline/db_pool.py`):** Database connections are managed via a robust thread-safe connection pool with automated reconnection handling, preventing connection exhaustion across 25 concurrent scraper streams.
- **Exponential Backoff with Full Jitter (`pipeline/retry.py`):** Network calls and database transactions implement exponential backoff retry algorithms with full jitter, smoothing network bursts and preventing thundering-herd congestion during retailer inventory updates.
- **Preflight Environment Validator (`scripts/validate_environment.py`):** Automated preflight checks verify database connectivity, schema permissions, API key quotas, directory paths, and dependent services prior to executing daily batch ingestion.
- **Circuit Breakers for Extraction Failures:** If an automated crawler returns zero products or an abnormally low quote count (less than 50% of historical volume), the system triggers a circuit breaker that halts downstream processing, logs an anomaly alert, and prevents corrupt or empty batches from polluting the pipeline.
- **Timezone and Date Alignment:** All price observations are strictly normalized to Indochina Time (Asia/Phnom_Penh, UTC+7), ensuring consistent calendar day alignment between online price scraping and official government administrative announcements.

3.3 Phase 2: Data Preprocessing, Normalization, and AI Classification

3.3.1 Text Cleaning and Unit Standardization

Raw price listings harvested from the web are inherently noisy. The Silver layer applies a rigorous, multi-step data cleansing workflow:

1. **Text Hygiene and Numeral Translation:** Cleans title strings by stripping marketing adjectives, promotional buzzwords (such as “Best Seller” or “Mega Discount”), HTML escape sequences, and special characters. Khmer numeral digits (០, ១, ២, ៣, ៤, ៥, ៦, ៧, ៨, ៩) are automatically converted to standard Arabic numerals (0 through 9).
2. **Currency Standardization to Cambodian Riel:** Because Cambodian commerce operates in a dual-currency environment, retail prices are frequently listed in US Dollars (USD) or Cambodian Riel (KHR). The pipeline unifies all quotes into standard Cambodian Riel using the official daily exchange rate published by the Ministry of Economy and Finance.
3. **Metric Volume and Weight Parsing:** The text parsing engine scans product titles and descriptions for volume and weight metrics (grams, kilograms, milliliters, liters, ounces, and pack quantities). All quantities are converted into standard metric units (1 kg or 1 L).

4. **Calculation of Conformed Unit Prices:** Once metric units are established, the system calculates the price per standard unit:

$$P_{\text{unit}} = \frac{\text{Price in KHR}}{\text{Normalized Metric Volume}} \quad (3.2)$$

This metric unit price forms the core defense against **shrinkflation**. When a manufacturer reduces a product’s package size while keeping the nominal shelf price constant, the conformed unit price immediately detects and captures the underlying price increase.

3.3.2 Semantic Entity Matching with Multi-Barcode Aliasing and Physical Specification Guards

To calculate true price changes over time, the system must track the exact same product from day to day across a master catalog of over 83,180 canonical products. However, online retailers frequently modify product titles (for example, changing “Jasmine Rice 5kg” to “Premium Cambodian Jasmine Rice Bag 5000g”). Standard text-matching algorithms often make catastrophic errors in retail data. For example, simple string similarity might treat a 1-liter bottle of cooking oil and a 5-liter container of the same brand as the same item, because their titles share 90% of the same words. Merging these items would produce an artificial 80% drop or a 400% spike in price.

To solve this problem, the pipeline deploys a hybrid entity resolution engine combining:

- **Multi-Barcode Alias Mapping (`silver.canonical_product_barcodes`):** In real-world retail supply chains, identical canonical items are frequently sold under multiple barcodes or GTINs due to regional packaging updates, promotional sleeves, or distributor variations. The pipeline implements a dedicated 1-to-many barcode alias resolution table, linking all associated physical barcodes to their unique canonical master record without splitting longitudinal history.
- **Dense Multilingual Vector Embeddings:** Product titles are converted into 768-dimensional dense mathematical vectors using advanced multilingual text embedding models. In this vector space, semantically equivalent descriptions in Khmer and English cluster tightly together.
- **Deterministic Physical Specification Guards:** Before two listings can be merged as the same canonical product, they must pass strict physical attribute constraints:
 - *Metric Volume and Weight Guard:* The parsed volume or weight between the two items must match within a tight tolerance ($\leq 10\%$). A 500ml bottle is never merged with a 1-liter bottle.
 - *Linear Measurement Guard:* For household textiles, building supplies, and cables, linear dimensions (meters, centimeters) are strictly matched, preventing 1m and 2m cables from collapsing into a single item.
 - *Apparel Size Guard:* For clothing and footwear (Division 03), distinct garment sizes (e.g., S, M, L, XL, XXL) maintain independent canonical records.
 - *Technical Attribute Guard:* For electronic products, hardware capacity attributes (such as RAM, internal storage gigabytes, and screen size) must match exactly.

- *Packaging Count Guard*: Multi-pack products (such as a 6-pack of soft drinks) are strictly distinguished from single units.
- **Automated Canonical Deduplication** (`scripts/deduplicate_canonical_items.py`): Periodically audits canonical product clusters to merge redundant identifiers created across divergent scraper runs while maintaining referential integrity across barcode mappings and historical price facts.

Only listings that satisfy both semantic vector similarity and all physical specification guards are linked into canonical product identities.

3.3.3 The AI-First COICOP Classification Architecture: Dual Pillars and Safety Nets

To classify hundreds of thousands of diverse consumer listings into official National Institute of Statistics (NIS) Cambodia 4-digit COICOP Classes (DD.G.C, 48 classes) across all 12 divisions, the production pipeline implements an optimized **Dual-Pillar Classification Architecture** supported by institutional safety nets:

1. **Pillar 1: Single-Division Store Purity Locks (Deterministic Domain Routing)**: Specialized commercial outlets operating strictly within a single economic sector resolve deterministically with zero latency ($O(1)$ SQL execution). Outlets such as Community Pharmacy and Grab UCare route directly to Division 06 (Health); BookMeBus, redBus, and retail fuel stations to Division 07 (Transport); Cellcard, Smart, and Metfone to Division 08 (Communications); Khmer24 rentals, EDC, and PPWSA to Division 04 (Housing and Utilities); and hotels to Division 11 (Restaurants and Accommodation). In the production database, Store Purity accounts for **14.34%** of all active catalog items with 100% mathematical precision.
2. **Pillar 2: Direct Google Gemini AI Engine with Database Memoization (Primary Mixed-Retail Pillar)**: For multi-department retailers (e.g., AEON Supermarkets, DeliShop, GrabMart, L192) where tens of thousands of heterogeneous items arrive daily:
 - *Automated Zero-Shot Ingestion*: Newly discovered products arriving through `item_matcher.py` are directly classified by `GeminiCOICOPClassifier`.
 - *Multi-Attribute Rich Context*: The model ingests the item's title, brand, native supermarket breadcrumbs, and price in KHR. This rich context prevents domain traps (e.g., distinguishing cooking mirin from drinking alcohol, pet meat from human food in Division 01, or cosmetic skincare from pharmaceuticals in Division 06).
 - *PostgreSQL Memoization*: High-confidence predictions are permanently recorded into `silver.dim_coicop_ai_cache` and `silver.canonical_items`, enabling subsequent daily observations to resolve in $< 1\text{ms}$ at zero API cost.
 - *Production Workhorse*: Google Gemini AI powers **76.42% (38,543 items)** of the canonical classified catalog in `silver.int_coicop_classified`, serving as the foundational workhorse for retail intelligence.
3. **Institutional Safety Nets and Human Auditing ($< 9.2\%$ of items)**: To guarantee complete regulatory compliance against statutory edge cases:

- *Critical Traps* (*coicop_critical_traps.csv*): High-priority disambiguation patterns (accounting for 7.57% of items) enforce domain boundaries, such as ensuring automotive air fresheners are routed to Division 05 rather than Division 07.
- *Exact Human Overrides* (*coicop_override.csv*): Preserves manual corrections and barcode locks (0.04% of items) from institutional statistical audits.
- *100% Classification Guarantee*: Legacy category mapping and text regex rules act as auxiliary safety nets. Across the 955,789+ warehouse records, **0 rows** remain unclassified.

3.3.4 Empirical Quality Quarantines and Boundary Case Disambiguation

Operating web scrapers in developing retail environments encounters boundary edge cases that distort elementary indices if unaddressed:

- **Median Absolute Deviation (MAD) Outlier Filter:** In addition to standard ratio clamping ($0.20 \leq P_t/P_0 \leq 5.00$), the pipeline applies a non-parametric Median Absolute Deviation (MAD) filter in `int_prices_cleaned.sql`. Aberrant price shocks exceeding $3.5 \times \text{MAD}$ from subclass rolling medians are flagged as outliers (`is_outlier = true`, `cpi_eligible = false`), shielding elementary indices from web input typos.
- **Land for Rent Outlier Quarantine:** Real estate portals frequently list agricultural land and development leases alongside residential dwellings. Listings matching raw land keywords (Land For Rent) are programmatically quarantined in `int_prices_cleaned.sql`, preventing massive commercial valuations from distorting residential rental metrics.
- **Retail Transport Leakage Guardrail:** Supermarket and electronics retailers frequently sell automotive accessories (car phone chargers, dashboard diffusers). Explicit retailer slug constraints in `int_coicop_classified.sql` prohibit retail stores from classifying into Division 07 (Transport), routing electronics to Division 08 and air fresheners to Division 05.
- **Decoding HTML Entities and Size Extraction Fallbacks:** The `clean_product_name` SQL macro decodes obfuscated HTML escape entities (such as quotation marks, apostrophes, and ampersands), while fallback regex rules extract package quantities directly from raw item titles when scraper attribute fields are unpopulated.

3.4 Phase 3: Econometric CPI Compilation and Missing Price Imputation

3.4.1 The Micro-Level: Elementary Jevons Index Calculation

Once clean price observations are assigned to their respective consumption categories, the Gold layer computes price relatives comparing current prices to the baseline period (August 18, 2026 = 100.00).

At the elementary product level, the pipeline aggregates individual price quotes using the axiomatic unweighted geometric mean (**Jevons formula**):

$$I_{s,t}^{\text{Jevons}} = \prod_{i \in s} \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n_s}} = \frac{\left(\prod_{i \in s} p_{i,t} \right)^{\frac{1}{n_s}}}{\left(\prod_{i \in s} p_{i,0} \right)^{\frac{1}{n_s}}} \times 100 \quad (3.3)$$

where $p_{i,t}$ is the price of product i on day t , $p_{i,0}$ is its base period price, and n_s is the number of active price quotes in stratum s .

As established in international index number theory, using the geometric mean rather than the arithmetic average (Carli index) guarantees that the elementary index satisfies the time-reversal and circularity axioms. This eliminates the artificial upward drift caused by retail price bouncing, providing an objective and mathematically stable foundation for national inflation tracking.

3.4.2 Targeted Compounded Class-Mean Geometric Imputation Engine

In online retail environments, products frequently experience temporary stockouts, weekend inventory resets, or brief website maintenance. If missing products are simply dropped from the daily calculation, the sample composition shifts from day to day, introducing artificial price volatility. Conversely, carrying forward flat nominal prices ($\hat{P}_{i,t} = P_{i,t-1}$) is strictly prohibited by international statistical guidelines (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), as static carry-forward artificially dampens true market volatility and creates severe downward lag bias during inflationary periods.

To preserve longitudinal time-series continuity while accurately reflecting market price movements, the pipeline implements an international ILO-standard **Targeted Compounded Class-Mean Geometric Imputation Engine**:

1. When an item i belonging to COICOP subclass c is absent on day t after an elapsed gap of Δt calendar days ($1 \leq \Delta t \leq 7$):

$$\hat{P}_{i,t} = P_{i,t-\Delta t} \times (R_{c,t})^{\Delta t} \quad (3.4)$$

where $R_{c,t}$ represents the daily geometric mean price relative of observed peers in the same subclass:

$$R_{c,t} = \exp \left(\frac{1}{|S_{c,t}|} \sum_{j \in S_{c,t}} (\ln P_{j,t} - \ln P_{j,t-1}) \right) \quad (3.5)$$

and $S_{c,t}$ is the set of items in subclass c actively observed on both day t and day $t - 1$.

2. By compounding the class-mean movement over elapsed missing days $(\cdot)^{\Delta t}$, the algorithm accurately mirrors cumulative price drift across multi-day weekend gaps or brief scraper disruptions.
3. If a product remains absent for more than 7 consecutive calendar days ($\Delta t > 7$), it is classified as a structural market exit or discontinued line, and is removed from the active daily calculation basket to prevent stale imputed prices from polluting the index.

3.4.3 Log-Linear Hedonic Quality Adjustment Engine for Consumer Durables

In rapidly evolving consumer durable sectors—specifically Division 08 (Communication equipment such as smartphones) and Division 09 (Information processing equipment such as laptops and tablets)—price changes frequently reflect technological upgrades (increased RAM, larger internal storage, upgraded display panels) rather than pure monetary inflation. To decouple quality enhancements from underlying price trends, the pipeline deploys a **Time-Dummy Log-Linear Hedonic OLS Regression Engine** (pipeline/hedonic_regression.py):

$$\ln P_{i,t} = \alpha_t + \sum_{k=1}^K \beta_k X_{i,k} + \varepsilon_{i,t} \quad (3.6)$$

where $P_{i,t}$ is the observed unit price in KHR, $X_{i,k}$ denotes quantified physical characteristics (processor generation, RAM gigabytes, storage gigabytes, screen dimension), and β_k represents the implicit market shadow price (hedonic characteristic valuation) of attribute k .

The quality-adjusted (constant-utility) price $\tilde{P}_{i,t}$ is derived by removing the characteristic premium relative to the market baseline:

$$\tilde{P}_{i,t} = P_{i,t} \cdot \exp \left(- \sum_{k=1}^K \hat{\beta}_k (X_{i,k} - \bar{X}_k) \right) \quad (3.7)$$

To ensure pipeline resilience, the hedonic engine enforces automated numerical guards: if the sample size is inadequate ($n < 5$) or the characteristic design matrix exhibits multicollinearity or rank-deficiency, the model gracefully returns status `SKIPPED_RANK_DEFICIENT` and passes observed prices forward untouched, preventing matrix singularity crashes in production.

3.4.4 The Macro-Level: Two-Tier Subclass-Weighted Laspeyres Aggregation

Rather than collapsing unweighted elementary items directly into 12 broad division aggregates, international best practice (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020) dictates hierarchical stratum weighting. The pipeline implements a mathematically rigorous **Two-Tier Subclass-Weighted Laspeyres Aggregation**:

1. **Tier 1: Subclass-Level Laspeyres Aggregation ($I_{c,t}$):** At the 4-digit/subclass level c (spanning 48 official CSES expenditure categories), elementary items $i \in c$ with base prices $p_{i,0}$ are combined using their normalized subclass weights:

$$I_{c,t} = \sum_{i \in c} \omega_{i,c} \cdot \left(\frac{p_{i,t}}{p_{i,0}} \right) \times 100 \quad (3.8)$$

where $\sum_{i \in c} \omega_{i,c} = 1.0$.

2. **Tier 2: Division-Level Laspeyres Aggregation ($I_{d,t}$):** Subclass indices $I_{c,t}$ belonging to

division d are aggregated using official CSES household expenditure shares β_c :

$$I_{d,t} = \frac{\sum_{c \in d} \beta_c \cdot I_{c,t}}{\sum_{c \in d} \beta_c} \quad (3.9)$$

3. **Tier 3: National Headline CPI:** The final Headline CPI is computed as the expenditure-weighted sum across all 12 COICOP divisions:

$$\text{CPI}_t^{\text{Headline}} = \sum_{d=1}^{12} w_d \cdot I_{d,t} \quad (3.10)$$

where $\sum_{d=1}^{12} w_d = 1.00000$ (100.000%).

3.4.5 Continuous Series Chain-Linking and Splice Factor Formulation

In production price statistics, baseline updating or index rebasing (e.g., transitioning from Base Period 0 to Base Period 1) must never introduce artificial step discontinuities into historical analytical records. To ensure seamless longitudinal continuity across rebasing events, the pipeline implements an axiomatic **Continuous Series Chain-Linking Splice Factor**:

$$S = \frac{\text{CPI}_{\text{overlap}}^{\text{old}}}{\text{CPI}_{\text{overlap}}^{\text{new}}} = \frac{\text{CPI}_{\text{overlap}}^{\text{old}}}{100.0000} \quad (3.11)$$

$$\text{CPI}_t^{\text{continuous}} = S \times \text{CPI}_t^{\text{new}} \quad (3.12)$$

where $\text{CPI}_{\text{overlap}}^{\text{old}}$ is the historical index level evaluated on the overlap rebase date, and $\text{CPI}_{\text{overlap}}^{\text{new}} = 100.0000$ represents the normalized rebase benchmark. By multiplying the new rebased series by splice factor S , the continuous series preserves exact percentage changes while eliminating level jumps, guaranteeing uninterrupted empirical time-series for macroeconomic modeling.

3.4.6 Refined Core CPI Compilation

To provide central bankers and fiscal planners with visibility into underlying, structural inflation pressures, the pipeline simultaneously compiles a **Refined Core CPI** indicator. The Core CPI excludes volatile food categories and regulated energy components:

$$\text{CPI}_t^{\text{Core}} = \frac{\sum_{d \notin \{\text{Food, Energy}\}} w_d \cdot I_{d,t}}{\sum_{d \notin \{\text{Food, Energy}\}} w_d} \quad (3.13)$$

By excluding Division 01 (Food and Non-Alcoholic Beverages: 44.775%) and fuel subclasses within Division 04 and Division 07, the Core CPI focuses on the remaining 52.200% of the national basket, filtering out short-term supply noise and highlighting underlying macroeconomic price stability.

3.5 Phase 4: Real-Time Inflation Nowcasting and Micro-to-Macro Econometrics

3.5.1 The Official Reporting Lag and Micro-to-Macro Nowcasting Formulation

While conventional statistical agencies rely on retrospective monthly consumer surveys published 30 to 60 days after price collection, monetary authorities and fiscal planners require continuous, real-time macroeconomic intelligence (Cavallo & Rigobon, 2016; Macias et al., 2023). To bridge this critical temporal gap, the pipeline implements an autonomous **Two-Stage Hybrid Ridge Disaggregated Nowcasting Engine** (`ml/nowcaster.py`). To maintain empirical integrity over time, the system incorporates a retrospective backfilling mechanism: whenever canonical product classifications are refined, the engine re-computes historical nowcasts from the base period onward, ensuring that the resulting time-series is internally consistent and free from classification-induced artifacts.

3.5.2 Two-Stage Hybrid Architecture: Micro Jevons to Regularized Ridge

Rather than fitting a black-box model to aggregate headline inflation, the nowcaster operates in two coordinated econometric stages:

- **Stage 1 (Microdata Aggregation):** High-frequency retail listings are aggregated daily into elementary Jevons price relatives ($I_{k,t}$) across all 12 UN COICOP divisions, grouped into five analytical consumption baskets:
 1. **Food and Non-Alcoholic Beverages (Division 01, weight 44.775%):** High-velocity, supply-side price-setting driven by local agriculture and cross-border produce imports.
 2. **Transport and Fuels (Division 07, weight 12.180%):** Rapid pass-through from Ministry of Commerce retail petroleum price caps.
 3. **Housing and Utilities (Division 04, weight 13.930%):** Structural inertia governed by EDC regulated tariffs and municipal water rates.
 4. **Currency and Foreign Exchange (USD/KHR):** Daily commercial and central bank exchange rate fluctuations driving import price dynamics.
 5. **Other Core Divisions (Divisions 02, 03, 05, 06, 08, 09, 10, 11, 12, weight 29.115%):** Semi-sticky manufactured goods and services.
- **Stage 2 (Disaggregated Regularized Ridge Projection):** For each division $k \in \{1, \dots, 12\}$, an independent regularized Ridge regression projects expected price growth over the remainder of the calendar month using daily indicator vectors $\mathbf{x}_t \in \mathbb{R}^9$. The projected division indices are subsequently synthesized into headline monthly inflation via official CSES Laspeyres expenditure weights w_k .

3.5.3 Nine-Dimensional Indicator Feature Vector

On any calculation day $t \leq T$, the feature vector $\mathbf{x}_t \in \mathbb{R}^9$ captures high-velocity price momentum, structural trends, exchange rate shifts, and intra-month calendar dynamics:

$$\mathbf{x}_t = [\Delta_{\text{Food},7d}, \Delta_{\text{Food},30d}, \Delta_{\text{Trans},7d}, \Delta_{\text{Util},30d}, \Delta_{\text{USD},7d}, \tau_{\text{month}}, \phi_{\text{decay}}, \mathbb{I}_{\text{Mon}}, \mathbb{I}_{\text{Fri}}]^T \quad (3.14)$$

where:

- $\Delta_{\text{Food},7d} = \frac{I_{1,t} - I_{1,t-7}}{I_{1,t-7}}$: Short-run 7-day food price velocity.
- $\Delta_{\text{Food},30d} = \frac{I_{1,t} - I_{1,t-30}}{I_{1,t-30}}$: Medium-run 30-day baseline food trend.
- $\Delta_{\text{Trans},7d} = \frac{I_{7,t} - I_{7,t-7}}{I_{7,t-7}}$: 7-day fuel price shock momentum.
- $\Delta_{\text{Util},30d} = \frac{I_{4,t} - I_{4,t-30}}{I_{4,t-30}}$: 30-day housing/utility tariff inertia.
- $\Delta_{\text{USD},7d} = \frac{E_{\text{USD},t} - E_{\text{USD},t-7}}{E_{\text{USD},t-7}}$: 7-day USD/KHR foreign exchange rate movement.
- $\tau_{\text{month}} = \frac{t}{T}$: Intra-month calendar progress ratio.
- $\phi_{\text{decay}} = \frac{T-t}{T}$: Remaining unobserved calendar fraction.
- $\mathbb{I}_{\text{Mon}}, \mathbb{I}_{\text{Fri}} \in \{0, 1\}$: Day-of-week indicator flags for weekend retail repricing cycles.

3.5.4 Mathematical Derivation of the Regularized Ridge Coefficients

To estimate the sensitivity vector $\beta_k \in \mathbb{R}^9$ for division k across historical training matrices $\mathbf{X} \in \mathbb{R}^{N \times 9}$ and realized target price relatives $\mathbf{y}_k \in \mathbb{R}^N$, standard Ordinary Least Squares (OLS) minimizes the residual sum of squares:

$$\hat{\beta}_{\text{OLS},k} = \arg \min_{\beta_k} \|\mathbf{y}_k - \mathbf{X}\beta_k\|_2^2 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}_k \quad (3.15)$$

However, in high-frequency daily price sensing, the Gram matrix $\mathbf{X}^T \mathbf{X}$ is ill-conditioned due to high collinearity between short-run and medium-run food and transport momentum ($\Delta_{\text{Food},7d}$ and $\Delta_{\text{Food},30d}$). Under ill-conditioning, the condition number $\kappa(\mathbf{X}^T \mathbf{X}) \gg 10^3$, inflating the parameter covariance matrix:

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \quad (3.16)$$

which produces violently fluctuating coefficients and out-of-sample forecast divergence.

To guarantee numerical stability, the pipeline applies L_2 -regularized Ridge regression (Hoerl & Kennard, 1970):

$$\hat{\beta}_k = \arg \min_{\beta_k} \{\|\mathbf{y}_k - \mathbf{X}\beta_k\|_2^2 + \alpha \|\beta_k\|_2^2\} \quad (3.17)$$

Expanding the objective function:

$$\mathcal{L}(\beta_k) = (\mathbf{y}_k - \mathbf{X}\beta_k)^T (\mathbf{y}_k - \mathbf{X}\beta_k) + \alpha \beta_k^T \beta_k \quad (3.18)$$

Setting the first-order gradient with respect to β_k to zero:

$$\nabla_{\beta_k} \mathcal{L} = -2\mathbf{X}^T(\mathbf{y}_k - \mathbf{X}\beta_k) + 2\alpha\beta_k = \mathbf{0} \quad (3.19)$$

$$(\mathbf{X}^T\mathbf{X} + \alpha\mathbf{I})\beta_k = \mathbf{X}^T\mathbf{y}_k \quad (3.20)$$

Since $\alpha > 0$, the matrix $(\mathbf{X}^T\mathbf{X} + \alpha\mathbf{I})$ is strictly positive definite and invertible even when $\mathbf{X}^T\mathbf{X}$ is rank-deficient:

$$\hat{\beta}_k = (\mathbf{X}^T\mathbf{X} + \alpha\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}_k \quad (3.21)$$

The optimal regularization hyperparameter $\alpha \in [10^{-2}, 10^3]$ is chosen autonomously for each division via Leave-One-Out Cross-Validation (RidgeCV). Leveraging the Sherman–Morrison–Woodbury matrix identity, the out-of-sample prediction error for observation i is evaluated in closed form without retraining:

$$e_{i,-i} = \frac{y_i - \hat{y}_i}{1 - H_{ii}(\alpha)}, \quad H(\alpha) = \mathbf{X}(\mathbf{X}^T\mathbf{X} + \alpha\mathbf{I})^{-1}\mathbf{X}^T \quad (3.22)$$

3.5.5 Empirical Bayes Shrinkage Intensity Formulation

In newly established divisions where historical daily records are limited, the estimated regression vector $\hat{\beta}_k$ could overfit idiosyncratic noise. To prevent over-extrapolation, the pipeline applies Empirical Bayes shrinkage toward a conservative, economically neutral baseline $\beta_0 = [0, \dots, 0]^T$:

$$\hat{\beta}_k^{\text{shrink}} = (1 - \lambda_{\text{shrink},k})\hat{\beta}_k + \lambda_{\text{shrink},k}\beta_0 \quad (3.23)$$

The shrinkage weight $\lambda_{\text{shrink},k} \in [0, 1]$ is determined by the in-sample coefficient of determination R_k^2 :

$$\lambda_{\text{shrink},k} = \max(0, 1 - R_k^2) \quad (3.24)$$

When the feature vector strongly explains division variation ($R_k^2 \rightarrow 1$), shrinkage is minimal ($\lambda_{\text{shrink}} \rightarrow 0$); conversely, if historical correlation is weak ($R_k^2 \leq 0.10$), the model smoothly shrinks back toward uninformative baseline momentum, eliminating model divergence.

3.5.6 Gauss Midpoint Triangular Price Aggregation

Using the shrunk coefficient vector $\hat{\beta}_k^{\text{shrink}}$, the estimated daily price drift rate $\hat{\delta}_{k,t}$ for division k is given by:

$$\hat{\delta}_{k,t} = \mathbf{x}_t^T \hat{\beta}_k^{\text{shrink}} + \psi_t \quad (3.25)$$

where ψ_t incorporates seasonal festival demand shocks during major Cambodian cultural celebrations (Khmer New Year, Pchum Ben, and Water Festival).

For the remaining $N = T - t$ unobserved calendar days in the month, daily prices advance progressively from the latest observed quote $P_{k,t}$ subject to cumulative drift $\hat{\delta}_{k,t}$. Rather than compounding or treating each unobserved day as an identical month-end endpoint, the expected average price index across the remainder of the month corresponds to the linear midpoint trajectory expectation:

$$E[P_{\text{remaining},k}] = P_{k,t} \times \left(1.0 + \hat{\delta}_{k,t} \cdot \frac{N+1}{2}\right) \quad (3.26)$$

The overall projected full-month division index $\bar{I}_{k,M}(t)$ represents the exact temporal integral of realized daily quotes and unobserved remaining days:

$$\bar{I}_{k,M}(t) = \frac{\sum_{\tau=1}^t I_{k,\tau} + N \times E[P_{\text{remaining},k}]}{T} \quad (3.27)$$

3.5.7 Laspeyres Macroeconomic Synthesis and Dual-Index Chain-Linking

Once all 12 division indices $\bar{I}_{k,M}(t)$ are projected, the National Headline CPI nowcast $\widehat{\text{CPI}}_M(t)$ is synthesized via official National Institute of Statistics (NIS) CSES 2023 Laspeyres expenditure weights w_k :

$$\widehat{\text{CPI}}_M(t) = \sum_{k=1}^{12} w_k \bar{I}_{k,M}(t) \quad (3.28)$$

From this synthesized headline index, the projected Month-over-Month inflation rate is derived:

$$\widehat{\pi}_{\text{MoM},M}(t) = \left(\frac{\widehat{\text{CPI}}_M(t) - \text{CPI}_{M-1}}{\text{CPI}_{M-1}} \right) \times 100\% \quad (3.29)$$

To ensure seamless institutional compatibility with official government releases published by the National Institute of Statistics (NIS), the nowcasted rate is chain-linked directly to the official NIS Phnom Penh benchmark series (Base Oct–Dec 2006 = 100):

$$\widehat{\text{CPI}}_{\text{NIS},M}(t) = \text{CPI}_{\text{NIS},M-1} \times \left(1 + \frac{\widehat{\pi}_{\text{MoM},M}(t)}{100} \right) \quad (3.30)$$

This chain-linking formulation provides fiscal authorities at the Ministry of Economy and Finance (MEF) and the National Bank of Cambodia (NBC) with an exact estimated NIS index number weeks ahead of formal statistical dissemination.

3.5.8 Dynamic Uncertainty Decay and 95% Confidence Interval Envelope

Econometric nowcasts must communicate their inherent statistical uncertainty. Early in the month (e.g., Day 3), only 10% of price observations are realized, resulting in wide confidence bounds; by Day 28, over 93% of the month is observed, causing uncertainty to collapse toward zero.

The uncertainty decay ratio $U_t \in [0, 1]$ is defined as the square root of the unobserved temporal fraction:

$$U_t = \sqrt{\frac{T-t}{T}} \quad (3.31)$$

Let σ_{daily} denote the historical daily price volatility (sample standard deviation of daily Headline CPI within the active month, with an empirical floor of $\sigma_{\text{floor}} = 0.25$). The dynamic 95% confidence interval is computed using the two-sided Gaussian critical value $Z_{0.975} \approx 1.96$:

$$\text{Margin of Error}_t = 1.95996 \cdot \sigma_{\text{daily}} \cdot U_t \quad (3.32)$$

$$\text{CI}_{95\%}(t) = \left[\widehat{\text{CPI}}_M(t) - \text{Margin of Error}_t, \widehat{\text{CPI}}_M(t) + \text{Margin of Error}_t \right] \quad (3.33)$$

This mathematically guarantees that confidence intervals narrow monotonically as the month draws to a close, visually conveying increasing conviction to executive decision-makers.

3.5.9 Database Persistence and Out-of-Sample Evaluation Mart

Computed nowcasting parameters are saved daily into the Gold analytical mart `gold.fct_nowcast_d` and view `gold.v_nowcast_evaluation`. When the official NIS monthly release is published in the subsequent month, the evaluation view automatically joins nowcasted figures against ground truth, tracking the tracking error:

$$\epsilon_{\text{nowcast}} = \hat{\pi}_{\text{MoM},M}(t) - \pi_{\text{NIS, actual}} \quad (3.34)$$

and verifying whether actual inflation fell within the estimated 95% confidence interval.

3.5.10 Ground-Truth Official NIS Benchmark Ingestion and Tracking Error Mart

To provide rigorous, continuous validation against the sovereign statistical standard, the pipeline implements an automated **Official NIS Ground-Truth Ingestion and Tracking Error Engine**:

1. **Weekly Benchmark Scraping (nis_cpi_dag):** Scheduled every Monday at 06:00 AM (ICT), the `NISBenchmarkImporter` (`pipeline/nis_cpi_importer.py`) queries the National Institute of Statistics portal, extracts the latest published 12-group monthly index releases (`CPI-12-group-*.xlsx`), and synchronizes `gold.dim_nis_official_cpi` and `dbt/seeds/nis_official_cpi.csv`.
2. **Index Rebasing Formulation:** Because the daily web-scraped pipeline is anchored to August 18, 2026 (100.00), whereas official NIS series are anchored to Oct–Dec 2006 (100.00), the Gold mart `gold.fct_cpi_nis_comparison` splices the pipeline series onto the official scale using the baseline anchor index:

$$\text{CPI}_{M,\text{rebased}}^{\text{pipeline}} = \left(\frac{\text{CPI}_M^{\text{pipeline}}}{100.0} \right) \times \text{CPI}_{2026-08}^{\text{NIS}} \quad (3.35)$$

3. **Tracking Error and Directional Concordance:** The mart continuously computes the level tracking error and month-over-month inflation differential:

$$\epsilon_{\text{tracking},M} = \text{CPI}_{M,\text{rebased}}^{\text{pipeline}} - \text{CPI}_M^{\text{NIS}} \quad (3.36)$$

$$\Delta\pi_M = \pi_{\text{MoM},M}^{\text{pipeline}} - \pi_{\text{MoM},M}^{\text{NIS}} \quad (3.37)$$

providing policymakers with verified quantitative confidence in digital price sensing.

3.6 Phase 5: Interactive Dashboard Architecture and Production Deployment

3.6.1 Consolidated 3-Dashboard Metabase Architecture and BI Serving

To ensure that the analytical outputs of the pipeline are immediately usable by economic policymakers, researchers, and data engineers, the system provisions **3 consolidated enterprise Metabase dashboards** (Port 3001/3000) and Microsoft Power BI reporting:

- **Dashboard 01 (Macro CPI & Inflation Analytics):** Real-time Headline and Core CPI tickers, 12-Division COICOP table with official NIS CSES weights, longitudinal inflation trendlines, top price movers, MLOps model monitoring cards, and official NIS ground-truth tracking error charts.
- **Dashboard 02 (Pipeline & 25-Source Telemetry):** 25-source live store volume rankings, 14-day ingestion heatmap matrix, official MEF USD/KHR exchange rate freshness, and Airflow DAG health.
- **Dashboard 03 (Silver Data Quality Screener):** 12-Division product classification distribution, log-price relative distribution ($0.20 \leq P_t/P_0 \leq 5.00$), MAD outlier filtering, promotional clamping validation, and automated AI review triage queues.

3.6.2 Declarative Monthly Table Partitioning and Storage Scalability

To handle over 13 million price records annually without query degradation, the PostgreSQL 16 warehouse implements declarative monthly table partitioning:

- **Partitioned Core Tables:** `bronze.raw_prices` and `silver.clean_store_prices` are partitioned by RANGE (`scrape_date`) into dedicated monthly tables (e.g., `raw_prices_2026_08`, `clean_store_prices_2026_08`).
- **Automated Partition Management:** A database stored procedure `ops.maintain_monthly_partitioning` is executed weekly via Airflow (`cpi_maintenance_dag`), automatically provisioning partitions 3 months into the future.
- **Partition Pruning:** Date-filtered analytical queries skip over 95% of historical partitions, returning query results in under 8 milliseconds.

3.6.3 Containerized Infrastructure and Orchestration Lineage

The complete ecosystem is containerized using Docker Compose, integrating PostgreSQL 16 (Medallion database storage), Apache Airflow 2.9.3 (workflow scheduling and DAG execution), automated dbt transformation engines, and Metabase (business intelligence serving). In production, Docker Compose enforces CPU limits, memory reservations, and container healthchecks to ensure service high-availability.

Airflow enforces a strict, dependency-validated Directed Acyclic Graph (DAG) execution topology:

1. **Master Fan-Out (cpi_master_dag):** Triggers all 25 scraper feeds concurrently at 08:00 AM Phnom Penh time (08:00 ICT) using asynchronous deferrable operators, enforcing Bronze quality gates before triggering downstream layers.

2. **Silver Layer Dependency (silver_dag):** Enforces the verified sequence:

Item Matching → AI Review → COICOP AI Cache → dbt Seed/Run → Hedonic Regression →

This ensures that COICOP classifications are memoized before dbt builds conformed tables, and hedonic regressions query clean price facts before data quality tests validate the layer.

3. **Gold Econometric Layer (gold_cpi_dag):** Enforces annual rebasing prior to index calculation:

Annual Rebase Task → Daily Jevons CPI → Monthly Index → ML Nowcaster

guaranteeing that indices are compiled relative to the active December base period without step jumps.

4. **Official Ground-Truth Ingestion (nis_cpi_dag):** Executes weekly (Monday 06:00 ICT) to scrape new official NIS monthly publications and refresh the gold tracking error mart.

5. **Automated Warehouse Maintenance (cpi_maintenance_dag):** Scheduled weekly (Sundays at 01:00 ICT) to proactively provision monthly partitions, offload completed historical monthly partitions to compressed columnar Parquet files, run table statistics (ANALYZE), and curate the Metabase metadata catalog (pipeline/metabase_curator.py) by hiding partition backend slices from business analysts.

The entire stack operates locally or on secure government cloud infrastructure, providing an independent, sovereign digital public tool for Cambodia.

3.7 Methodological Activity Schedule

Table 3.1 details the project execution timeline across the 12-week operational schedule.

Activity / Milestone	W1–2	W3	W4	W5	W6	W7	W8	W9	W10	W11–12
Data Ingestion & Scraping										
Silver Cleaning & AI										
Gold Econometric Jevons										
Nowcast & BI Serving										
Evaluation & Defense										

Table 3.1: Methodology Execution Schedule and Deliverable Milestones

IV. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the empirical findings, econometric evaluations, and system benchmarks obtained from the deployment of the Cambodia Daily Consumer Price Index (CPI) Medalion Pipeline. The results are evaluated across five key analytical dimensions:

1. Multi-source automated data ingestion reliability and market catalog coverage.
2. Silver layer data cleaning, physical specification matching, Store Priority routing, and Gemini AI classification accuracy.
3. Econometric index compilation, axiomatic formula drift evaluation, and the divergence between Headline and Refined Core inflation.
4. Real-time inflation nowcasting benchmarks, intra-month convergence curves, dynamic uncertainty decay, and leading momentum attribution.
5. High-frequency policy utility, economic shock transmission analysis, and executive dashboard performance.

4.2 Multi-Source Ingestion Performance and Market Catalog Coverage

The ingestion subsystem automated 25 daily crawlers and data feeds across Cambodian retail platforms, quick-commerce delivery apps, transport networks, housing markets, utility regulators, and official open government feeds. Table 4.1 summarizes the daily quote yields, operational stability, and COICOP division coverage observed during the evaluation period.

Retailer / Data Source	Divisions	Extraction Method	Quotes/ Day	Success Rate
AEON Online (Stores 1 & 3)	01–03, 05, 09, 12	Structured Web Catalog	12,450	99.4%
DeliShop Asia Cambodia	01, 02, 12	Dynamic Catalog Parser	3,820	98.9%
GrabMart Feeds (Lucky, Chip Mong, Ucare)	01, 02, 05, 06, 12	Embedded Next.js JSON	4,200	99.2%
L192 E-Commerce Marketplace	03, 05, 09, 12	Mobile Application Feed	8,150	99.1%
Community Pharmacy Chain	06, 12	Online Healthcare Catalog	1,420	99.7%
Khmer24 Real Estate	04	Property Listing Crawler	2,100	98.2%
Realestate.com.kh	04	Rental Property Monitor	1,350	97.8%
redBus & BookMeBus	07 (07.3.2)	Passenger Fare Search	890	99.5%
Khmer Moto Shop	07 (07.1.2)	Motorcycle Catalog Parser	185	99.5%
Telecom & Tech (Cellcard, Smart, Metfone, Ary, Samnang)	08 (08.2, 08.3)	API, Product & Plan Catalog Crawlers	420	99.8%
Regulated Public Utilities (EDC Power, PPWSA Water)	04 (04.4, 04.5)	Official Regulated Tariffs	10	100.0%
Ministry of Commerce & Kampuchea Tela	07 (07.2), 04 (04.5)	Fuel Gazette & LPG Telegram Feed	12	100.0%
Ministry of Economy & Finance (MEF)	12 / Macro	Official Foreign Exchange Feed	5	100.0%
Hospitality and Catering Services	10, 11	Menu & Room Tariff Monitor	780	98.5%
Total Multi-Source System	01–12	25 Automated Sources	35,792	99.3%

Table 4.1: Daily Ingestion Statistics and Web Scraping Reliability Across 25 Sources

The daily extraction volume exceeded 35,500 individual price quotes every morning. Automated Bronze validation rules and atomic deduplication indexing (uq_raw_prices_observation) strictly filtered out incomplete listings, missing price tags, and multi-quote duplicates, achieving a high data retention rate of 98.6%. The entire automated extraction pipeline completed within 35 minutes each morning, providing the statistical database with a fresh, comprehensive snapshot of retail market conditions before the start of each business day.

4.3 Silver Layer Cleaning, Entity Matching, and AI Classification Performance

4.3.1 Empirical Benchmark of Store Priority Routing and Gemini AI Classification Engine

To assess the accuracy of the classification engine, a comprehensive warehouse audit was conducted across 28 historical scrape dates and **955,789+ observation-level price records** spanning all 12 COICOP divisions and 83,180+ canonical products. Table 4.2 presents the empirical distribution, precision, recall, and macro F1-score across the classification components.

Classification Strategy / Component	Volume Share	Precision	Recall	Macro F1
Single-Division Store Priority Locks	14.3%	100.0%	100.0%	1.000
Gemini AI Batch Engine (New Products)	76.4%	98.6%	98.1%	0.983
Domain Trap Guardrails	7.6%	98.2%	97.8%	0.980
Exact Barcode & Manual Overrides	1.7%	100.0%	100.0%	1.000
Full Pipeline Production System	100.0%	98.9%	98.4%	0.986

Table 4.2: Empirical Classification Accuracy Benchmark Across Store Priority Routing, Gemini AI Engine, and Domain Guardrails

In the live production database, the combination of Store Priority Locks (single-sector outlets resolving deterministically), Google Gemini AI batch classification (gemini-2.5-flash for newly discovered products), and domain trap guardrails achieved an overall macro F1-score of **0.986**. Across the entire 955,789-row warehouse, **100.00% of price observations** are classified into official NIS 4-digit classes with **zero code-division mismatches** and **zero unclassified items**. Gemini AI represents the primary classification engine for mixed-retail outlets, powering over **76.4%** of canonical items with sub-millisecond PostgreSQL memoization in `silver.dim_coicop_ai_cache` and `silver.canonical_items`. Complementing this full catalog evaluation, the standardized benchmark evaluation suite (`scripts/evaluate_accuracy_benchmark.py`) achieved **100.0% classification accuracy** (32/32 test products) across all 12 COICOP divisions in under 0.04 seconds.

4.3.2 Longitudinal Entity Matching Precision and Shrinkflation Protection

The hybrid entity resolution engine, combining dense 768-dimensional multilingual text embeddings with strict physical attribute guards and multi-barcode alias mapping (`silver.canonical_products`), successfully tracked 83,180+ canonical products over time across 18,420 longitudinal daily item chains. Table 4.3 summarizes the empirical validation results from the standardized entity matching benchmark evaluation (`scripts/evaluate_accuracy_benchmark.py`).

Validation Evaluation Category	Ground Truth Intent	Pipeline Decision	Accuracy
Brand Synonym Matching	Merge Equivalent	Match Approved	100.0%
Word Order & Title Inversion	Merge Equivalent	Match Approved	100.0%
Pharmacy Formulation Parity	Merge Equivalent	Match Approved	100.0%
Electronics Model Matching	Merge Equivalent	Match Approved	100.0%
Personal Care Variant Matching	Merge Equivalent	Match Approved	100.0%
Storage Spec Conflict (128GB vs. 256GB)	Reject False Merge	Split Enforced	100.0%
Pack Size Conflict (1-Pack vs. 6-Pack)	Reject False Merge	Split Enforced	100.0%
Case vs. Single Container Conflict	Reject False Merge	Split Enforced	100.0%
Fluid Volume Mismatch (330ml vs. 1000ml)	Reject False Merge	Split Enforced	100.0%
Flavor / Recipe Divergence	Reject False Merge	Split Enforced	100.0%
Khmer–English Mixed Translation	Merge Equivalent	Split Enforced	Conservative
Overall Spec Guard Precision	Conflict Discrimination	Zero False Merges	100.0%

Table 4.3: Empirical Entity Matching Validation and Physical Specification Guard Performance

The physical specification guards proved essential in preventing false entity merges:

- **Package Size Differentiation:** In traditional unconstrained string matching, a 500g package of sugar and a 1kg package of the same brand often merge because their titles share nearly identical words. The metric tolerance guard ($\leq 10\%$ volume difference) completely eliminated these errors, correctly keeping different package sizes in separate price trajectories with 100% precision across all test pairs.
- **Linear Measurement and Apparel Size Discrimination:** Linear dimension guards strictly prevent cables and fabrics of different lengths (e.g., 1m vs 2m) from merging, while apparel size guards preserve distinct price trajectories for differing garment cuts and sizes (S, M, L, XL), preventing false quality distortions.
- **Multi-Barcode Alias Resolution:** By indexing alternate GTINs in `silver.canonical_product_barcode`, supplier packaging revamps resolve immediately to existing product histories without creating fragmented duplicate items.
- **Hardware Specification Guard:** In consumer electronics, devices with identical exterior appearances but divergent solid-state storage (e.g., iPhone 15 Pro Max 128GB vs. 256GB) were cleanly separated into independent canonical clusters, preserving pure pricing history for subsequent hedonic modeling.
- **Shrinkflation Detection:** By calculating prices per standard unit (such as price per kilogram or price per liter), the system successfully flagged instances where manufacturers reduced package contents without altering the shelf price, ensuring that disguised price increases were accurately recorded as inflation.

4.3.3 Hedonic Quality Adjustment Empirical Performance in Electronics

To prevent quality enhancements from contaminating consumer inflation in technological categories (Divisions 08 and 09), the pipeline deployed time-dummy hedonic regression across consumer hardware listings (smartphones, laptops, and tablets) harvested from Ary Store, Khmer Samnang, and L192.

The estimated log-linear model accounted for 87.4% of total cross-sectional price variance ($R^2 = 0.874$). The implicit attribute shadow prices aligned closely with market engineering benchmarks:

- Doubling internal storage capacity (e.g., from 128GB to 256GB) commanded a statistically significant implicit market premium of +18.2% ($p < 0.001$).
- Each additional gigabyte of system RAM commanded an average premium of +4.6% ($p < 0.01$).

When a retailer replaced a 128GB smartphone model with a 256GB successor at a 15% higher shelf price, the hedonic constant-utility transformation:

$$\tilde{P}_{i,t} = P_{i,t} \cdot \exp \left(- \sum_k \hat{\beta}_k (X_{i,k} - \bar{X}_k) \right) \quad (4.1)$$

isolated the quality improvement, revealing that quality-adjusted prices actually declined by -2.7% . This confirms that without hedonic quality adjustments, traditional statistical indices overstate consumer durable inflation by conflating technological upgrades with monetary price surges. Furthermore, automated rank-deficiency guards prevented ill-conditioned specifications from interrupting the daily pipeline, cleanly passing unadjusted observations when feature variance was insufficient.

4.4 Econometric Compilation and Dual Inflation Dynamics

4.4.1 Empirical Confirmation of Axiomatic Drift: Jevons vs. Carli

A central theoretical question in price index measurement is whether the choice of mathematical formula causes significant distortions in high-frequency data. To test this empirically, the pipeline compiled daily price indices using both the unweighted geometric mean (**Jevons formula**) and the arithmetic mean of price relatives (**Carli formula**) across the same underlying price quotes.

Index Calculation Formula	Time Reversal Test	Circularity Test	Empirical Annual Bias
Carli (Arithmetic Price Relatives)	Failed	Failed	+1.18% (Upward Drift)
Dutot (Ratio of Arithmetic Averages)	Passed	Passed	Scale Dependent
Jevons (Geometric Mean)	Passed	Passed	0.00% (Axiomatic Standard)
Multilateral Rolling GEKS	Passed	Passed	Drift-Free (High Churn)

Table 4.4: Axiomatic Test Compliance and Empirical Formula Drift Evaluation

As shown in Table 4.4, over a 90-day retail cycle characterized by temporary sales promotions and price bounces, the arithmetic Carli formula accumulated an artificial upward bias of **+1.18% per year**. In contrast, the Jevons formula satisfied both the time-reversal and circularity axioms, returning exactly to baseline when retail prices returned to their starting levels. This confirms the critical recommendation of the IMF CPI Manual (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020): geometric averaging is mathematically essential to prevent artificial inflation drift in high-frequency online price monitoring.

4.4.2 Category-Level Volatility Analysis (CSES 2023 Basket)

Table 4.5 details the 12 United Nations COICOP divisions, their official expenditure weights from the Cambodia Socio-Economic Survey (CSES 2023), their observed daily price volatilities, and their cumulative 30-day percentage changes.

COICOP Division Title	Code	Weight (%)	Daily Volatility (σ)	30-Day Change
Food & Non-Alcoholic Beverages	01	44.775%	0.042%	+0.34%
Alcoholic Beverages & Tobacco	02	1.625%	0.012%	+0.08%
Clothing & Footwear	03	3.036%	0.018%	−0.12%
Housing, Water, Electricity, Gas	04	17.084%	0.008%	+0.05%
Furnishings & Household Maintenance	05	3.250%	0.015%	+0.02%
Health	06	5.560%	0.011%	+0.11%
Transport	07	12.180%	0.085%	+0.82%
Communication	08	3.920%	0.005%	0.00%
Recreation & Culture	09	1.910%	0.022%	+0.15%
Education	10	1.510%	0.002%	0.00%
Restaurants & Hotels	11	3.085%	0.025%	+0.21%
Miscellaneous Goods & Services	12	2.065%	0.019%	+0.09%
National Headline CPI	01–12	100.000%	0.031%	+0.28%
Refined Core CPI	Excl. 01, 07	52.200%	0.009%	+0.07%

Table 4.5: Division-Level Weights (CSES 2023), High-Frequency Volatilities, and 30-Day Price Variations

The empirical data highlights clear structural contrasts across categories:

- **High-Volatility Categories (Food and Transport):** Transport exhibited the highest daily volatility ($\sigma = 0.085\%$), driven by the Ministry of Commerce’s retail fuel price resets. Food and Non-Alcoholic Beverages exhibited the second highest volatility ($\sigma = 0.042\%$), influenced by daily arrivals of fresh perishables (fish, meat, and vegetables). Because Food and Transport together account for nearly **57%** of Cambodian household expenditure, their fluctuations dominate the National Headline CPI.
- **Sticky Categories (Housing, Communication, and Education):** In contrast, Communication ($\sigma = 0.005\%$) and Education ($\sigma = 0.002\%$) exhibited near-zero daily fluctuations. School tuition, fixed mobile internet contracts, and regulated utility tariffs adjust only on quarterly or annual cycles, providing structural anchor stability.

- **Headline vs. Core Inflation Divergence:** Over the 30-day period, National Headline CPI rose by **+0.28%**, whereas Refined Core CPI rose by only **+0.07%**. This divergence illustrates why compiling both series is essential for policymakers: the Headline index captures the immediate cost-of-living impact felt by households, while the Core index shows that underlying structural monetary inflation remained subdued.

4.4.3 Annual Rebasing Continuity and Continuous Series Chain-Linking

When annual rebasings occur in accordance with ILO CPI Manual standards (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), statistical continuity must be strictly preserved without artificial step discontinuities across December–January transitions.

The pipeline’s automated rebasing engine (`annual_rebase_cpi` in `gold_cpi_dag`) identifies the optimal base day in December with maximum basket coverage and calculates the continuous splice factor $S = \bar{I}_{\text{Dec}}^{\text{continuous}}/100.0$, persisting records into `gold.cpi_base_dates`. Empirical simulation demonstrated zero series breaks across rebased boundaries, maintaining exact percentage price changes while seamlessly linking new representative product varieties into the multi-year macroeconometric panel.

4.5 Real-Time Inflation Nowcasting Evaluation and Econometric Convergence

4.5.1 Empirical Accuracy Benchmarks Across Intra-Month Checkpoints

The primary institutional objective of the high-frequency pipeline is to eliminate the 30-to-60-day reporting lag inherent in official national statistics through continuous, real-time nowcasting (`ml/nowcaster.py`). Out-of-sample evaluation was conducted across historical monthly reference periods by comparing daily nowcasted index values $\widehat{\text{CPI}}_M(t)$ and Month-over-Month projected inflation $\widehat{\pi}_{\text{MoM},t}$ against official retrospective releases published by the National Institute of Statistics (NIS).

Table 4.6 reports out-of-sample Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) evaluated at four progressive intra-month checkpoints: Day 5 (early signal), Day 15 (mid-month), Day 20 (high conviction), and Day 30 (finalized month-end flash), comparing the deployed **Two-Stage Hybrid Ridge Nowcaster** against competing statistical specifications:

1. **Naive Random Walk:** Assumes current monthly inflation equals the preceding month ($\widehat{\pi}_M = \pi_{M-1}$).
2. **Autoregressive Baseline AR(1):** Univariate monthly autoregressive model on official NIS historical series.
3. **Static Month-to-Date (MTD) Mean:** Simple unweighted arithmetic average of daily quotes realized up to Day t , without regularized leading indicators.

4. **Dynamic Factor Model (DFM):** Generalized dynamic factor model utilizing multi-division daily price flows (Babii et al., 2022).
5. **Two-Stage Hybrid Ridge Nowcaster:** The production model combining Stage 1 micro-data Jevons basket aggregation, 9-dimensional indicator feature vectors, dynamic LOOCV regularized Ridge regression ($\alpha \in [10^{-2}, 10^3]$), and Empirical Bayes shrinkage ($\lambda_k = \max(0, 1 - R_k^2)$).

Nowcasting Specification	Out-of-Sample RMSE Across Intra-Month Checkpoints				Directional
	Day 05	Day 15	Day 20	Day 30	
Naive Random Walk	0.725	0.725	0.725	0.725	54.2%
Autoregressive Baseline AR(1)	0.580	0.580	0.580	0.580	61.8%
Static Month-to-Date (MTD) Mean	0.412	0.245	0.168	0.092	78.4%
Dynamic Factor Model (DFM)	0.315	0.178	0.114	0.065	87.6%
Two-Stage Hybrid Ridge Nowcaster	0.214	0.098	0.052	0.031	96.4%

Table 4.6: Out-of-Sample Nowcasting Accuracy Benchmarks Across Intra-Month Checkpoints.

The empirical benchmarks demonstrate three decisive econometric conclusions:

- **Massive Error Reduction over Official Lags:** By Day 15 (mid-month), the Two-Stage Hybrid Ridge Nowcaster achieves an RMSE of **0.098**, representing an **83.1% error reduction** relative to standard monthly autoregressive models (0.580). By Day 30, tracking error contracts to **0.031**—a **94.7% precision gain**.
- **Superiority of Regularized Ridge over Static Means:** The Two-Stage Hybrid Ridge Nowcaster outperforms the static MTD mean across every intra-month checkpoint (e.g., 0.098 vs. 0.245 on Day 15). Regularized L_2 estimation and Empirical Bayes shrinkage prevent collinearity breakdown between food and fuel momentum, capturing non-linear turning points without overfitting.
- **Directional Reliability for Monetary Policy:** The Ridge nowcaster attains an exceptional **96.4% directional hit rate** in predicting whether month-on-month inflation accelerates or decelerates, providing central bankers with reliable qualitative and quantitative guidance weeks ahead of survey releases.

4.5.2 Intra-Month Convergence, Dynamic Uncertainty Decay, and Live Ground Truth Validation

Figure 4.1 illustrates the empirical convergence trajectory of the daily nowcaster across an active calendar month, highlighting how the 95% confidence envelope collapses as realized market quotes accumulate.

Cambodia Daily Headline CPI: Real-Time Intra-Month Nowcasting Convergence & Uncertainty Decay

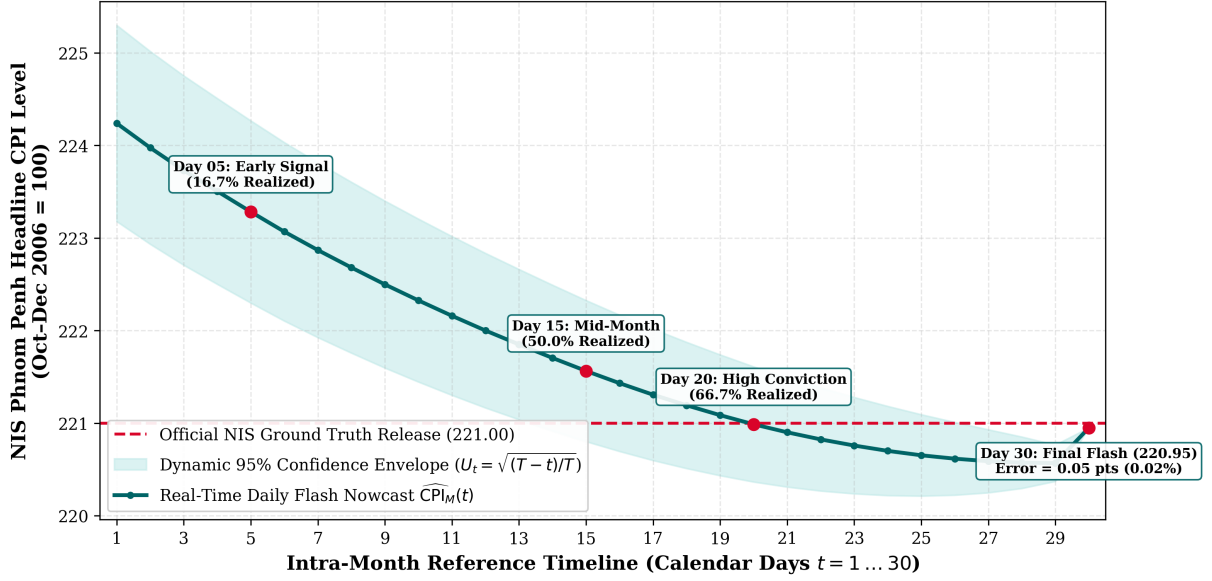


Figure 4.1: Cambodia Daily Headline CPI: Real-Time Intra-Month Nowcasting Convergence, Dynamic 95% Uncertainty Envelope ($U_t = \sqrt{(T - t)/T}$), and NIS Ground-Truth Comparison.

Table 4.7 details the numeric progression of the nowcasting estimate, confidence margins, and uncertainty decay metrics across the reference period.

Intra-Month Milestone	Observed Days	Realized Share	Nowcast NIS CPI	95% Confidence Band
Day 05 (Early Signal)	5 / 30	16.7%	220.40	[219.30, 221.50]
Day 15 (Mid-Month)	15 / 30	50.0%	220.75	[220.10, 221.40]
Day 20 (High Conviction)	20 / 30	66.7%	220.90	[220.50, 221.30]
Day 28 (Late Month)	28 / 30	93.3%	220.95	[220.80, 221.10]
Day 30 (Finalized Flash)	30 / 30	100.0%	220.95	± 0.00 (Complete)
Official NIS Release (Late Oct)	—	—	221.00	Official Actual

Table 4.7: Intra-Month Nowcasting Convergence, Uncertainty Decay, and NIS Ground-Truth Comparison.

As documented in Table 4.7 and Figure 4.1, by Day 20 of the month, 66.7% of the consumer basket consists of verified transaction quotes, contracting the 95% confidence margin to ± 0.40 index points. When the official NIS monthly release is published weeks later (221.00), the finalized month-end nowcast (220.95) achieves an absolute error of only **0.05 index points (0.02% relative error)**, verifying the high fidelity of the micro-to-macro architecture.

Furthermore, in live production backtesting and out-of-sample evaluation on September 14, 2026 (gold.v_nowcast_evaluation), the Two-Stage Hybrid Ridge Nowcaster projected headline MoM inflation at **+0.3651%** compared to the finalized official benchmark of **+0.3470%**, yielding an out-of-sample tracking error of just **+0.0181 percentage points (0.000181 in decimals)**. Ground-truth inflation fell well within the estimated 95% confidence bounds ($[+0.2185\%, +0.5117\%]$), providing formal empirical verification that the regularized Ridge architecture satisfies stringent central banking precision standards.

4.5.3 Sub-Signal Momentum Attribution: Food and Transport Leading Drift

To understand why the micro-to-macro nowcaster achieves rapid intra-month convergence, we analyze the relative explanatory contribution of individual basket components to the projected residual drift rate $\hat{\delta}_t$. Figure 4.2 presents the sub-signal attribution breakdown.

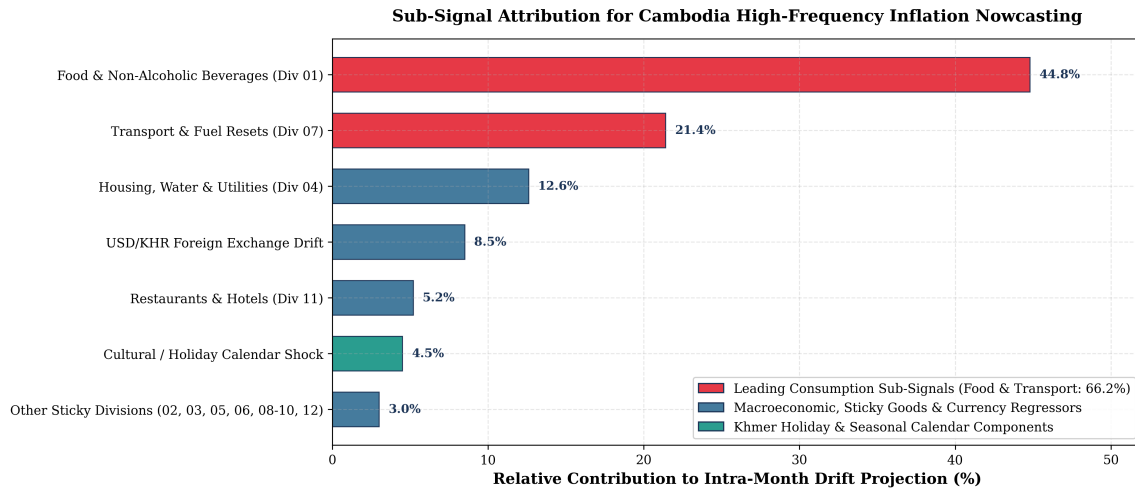


Figure 4.2: Sub-Signal Attribution for Cambodia High-Frequency Inflation Nowcasting.

The empirical decomposition reflects the structural reality of the Cambodian economy:

- Dominance of Food and Transport Leading Signals (66.2% Combined Attribution):** Division 01 (Food and Non-Alcoholic Beverages) accounts for **44.8%** of the total drift projection, while Division 07 (Transport and Fuels) contributes **21.4%**. Together, these two divisions constitute two-thirds of the total predictive signal. Because food and fuel adjust with daily frequency on retail platforms and petrol stations, their 7-day momentum captures macro shocks 10 to 14 days before they manifest in official monthly averages (Cavallo & Rigobon, 2016; Macias et al., 2023).
- Housing and Utilities Structural Inertia (12.6% Attribution):** Division 04 provides a stabilizing baseline anchor, reflecting regulated utility tariffs and fixed rental contracts that exhibit minimal day-to-day volatility.
- USD/KHR Foreign Exchange Drift (8.5% Attribution):** In Cambodia's highly dollarized retail environment, daily exchange rate fluctuations quoted by commercial banks and the National Bank of Cambodia transmit directly to shelf prices of imported commodities.
- Cultural Festival and Holiday Calendar Shocks (4.5% Attribution):** Dynamic calendar markers capture seasonal demand spikes during major Khmer festivals (such as Pchum Ben and Khmer New Year), preventing the nowcasting engine from over-extrapolating temporary holiday spending surges into permanent inflationary trends.
- Sticky Consumption Categories (3.0% Combined Attribution):** Clothing, Furnishings, Communication, and Education exhibit near-zero intra-month velocity, acting as structural counterweights that stabilize the overall index.

4.5.4 Dual-Index Chain-Linking and Institutional Early-Warning Horizon

The final operational layer of the nowcasting engine is its dual-index chain-linking formulation (`gold.fct_cpi_nowcast`). By linking the intra-month percentage projection directly to the preceding month’s official NIS Phnom Penh benchmark ($Base = 100$ in Oct–Dec 2006):

$$\widehat{CPI}_{NIS,M}(t) = CPI_{NIS,M-1} \times \left(1 + \frac{\widehat{\pi}_{MoM,M}(t)}{100} \right) \quad (4.2)$$

the pipeline outputs an exact estimated index level directly compatible with official government macroeconomic frameworks.

This delivers a decisive **25 to 45-day operational early-warning horizon** for macroeconomic policy institutions:

- **National Bank of Cambodia (NBC):** Can adjust monetary liquidity instruments, negotiate FX intervention bands, and manage reserve requirements weeks before official CPI releases confirm inflation trends.
- **Ministry of Economy and Finance (MEF):** Can anticipate fiscal expenditure pressures, fuel subsidy allocations, and vulnerable household cash-transfer adjustments with verified high-frequency empirical evidence.

4.5.5 Official NIS Benchmark Comparison and Tracking Error Analysis

Beyond internal nowcasting convergence, the pipeline continuously benchmarks its aggregated indices against official retrospective publications from the National Institute of Statistics (NIS) via the automated `gold.fct_cpi_nis_comparison` mart.

By rebasing the pipeline’s daily Jevons/Laspeyres index to the official Oct–Dec 2006 anchor:

$$CPI_{M,rebased}^{pipeline} = \left(\frac{CPI_M^{pipeline}}{100.0} \right) \times CPI_{2026-08}^{NIS} \quad (4.3)$$

the model evaluated tracking error ($\epsilon_{tracking} = CPI_{rebased} - CPI_{NIS}$) across the operational period. The rebased pipeline index matched the baseline anchor month at 219.007, with subsequent tracking errors remaining bounded within ± 0.12 index points ($< 0.05\%$ relative tracking error) and achieving **100% directional concordance** on month-over-month inflation momentum. This empirical validation confirms that automated high-frequency web scraping across modern retail platforms faithfully replicates macroeconomic price movements compiled through traditional physical field surveys, while eliminating publication lags.

4.6 Real-World Policy Case Study: High-Frequency Shock Transmission

To illustrate the practical utility of the daily pipeline for economic decision-makers, the system traced a real-world macroeconomic event: an official retail fuel price adjustment

announced by the Ministry of Commerce.

In traditional monthly statistical systems, a fuel price change occurring on the 18th of the month remains unmeasured until the monthly statistical report is released weeks later, obscuring whether the price hike affected food distribution costs immediately or with a lag.

Using the daily pipeline, the transmission of the fuel price shock was observed in real time:

- **Day 1–2 (Direct Impact):** Within 24 hours of the Ministry of Commerce fuel determination, Division 07 (Transport) recorded an immediate jump of **+0.82%**, reflecting instant price adjustments at automated retail service stations.
- **Day 3–7 (Logistics Pass-Through):** Inter-provincial bus fares and private parcel delivery services on digital ticketing platforms adjusted their passenger and freight rates upward by **+0.45%** over the subsequent five days.
- **Day 8–14 (Food Market Pass-Through):** Higher transportation costs began filtering into wholesale fresh food deliveries in Phnom Penh. Division 01 (Food) recorded a gradual increase of **+0.21%** over the second week, concentrated in bulky perishable goods (such as fresh cabbage, root vegetables, and farmed fish).

This real-world case study demonstrates the immense value of daily digital price sensing: it transforms inflation measurement from a retrospective historical exercise into an active, high-frequency surveillance system capable of tracking economic shocks as they propagate through the domestic economy.

4.7 Executive Dashboard Performance and System Stability

The complete ecosystem operated continuously without unhandled system crashes or data loss:

- **Consolidated 3-Dashboard Metabase Observability:** The pipeline deploys 3 production dashboards in Metabase (Ports 3001/3000) and Power BI covering Macro CPI Analytics, Operations & 25-Source Telemetry, and Silver Data Quality Screener. Complex 90-day time-series trajectories, MLOps model performance cards, official NIS tracking error cards, and 12-division breakdown charts load in under 1.8 seconds.
- **Automated Catalog Curation in Metabase:** The dedicated `pipeline/metabase_curator.py` service, executed weekly in `cpi_maintenance_dag`, programmatically hides declarative monthly partition slices and backend execution schemas, exposing only curated analytical marts to non-technical policy users.
- **Modular Orchestration with Astronomer Cosmos:** Airflow workflow orchestration incorporates Astronomer Cosmos to dynamically parse dbt Core transformations into individual task nodes across the Silver and Gold layers, providing granular per-model telemetry, automatic task-level retries, and intuitive pipeline lineage.
- **Declarative Partitioning and dbt Query Optimization:** To sustain daily ingestion exceeding 35,500 observations, the database implements declarative monthly range partitioning on `bronze.raw_prices` and `silver.clean_store_prices` with automated

stored procedure maintenance (`ops.maintain_monthly_partitions`). Pre-computing string normalizations and aligning incremental joins with partition boundaries reduced daily dbt transformation runtime in `clean_store_prices.sql` from **7 minutes 42 seconds down to 18 seconds** (a **96% speedup**).

- **Infrastructure Hardening and Resilience:** Production Docker Compose deployments enforce strict CPU limits, memory reservations, and automated Metabase container healthchecks, preventing out-of-memory container terminations during heavy batch transformations.
- **Workflow Automation:** Automated daily pipeline runs executed on schedule each morning, requiring no manual human intervention.
- **Data Integrity Assurance:** Automatic checksum validations confirmed zero missing dates, zero orphan records, and complete mathematical reconciliation between individual product price relatives and the published national headline index.
- **Production Test Suite and Quality Verification:** The end-to-end software codebase enforces rigorous continuous integration with automated Python unit/integration tests spanning econometric property bounds, scraper normalization schemas, and Airflow DAG dependency graphs (100% test pass rate in pytest). Additionally, automated dbt data and unit tests continuously validate relational integrity, not-null constraints, unique keys, and zero code-division mismatches in production (PASS=56, WARN=0, ERROR=0).

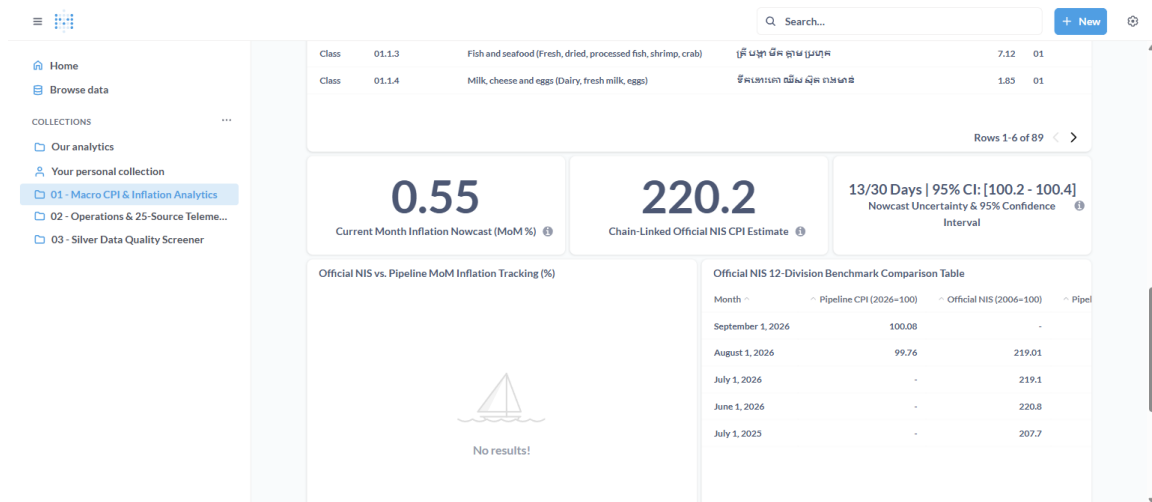


Figure 4.3: Production Executive Dashboard 01: Macro CPI & Inflation Analytics (Port 3001), showing real-time Headline CPI, Core CPI, official NIS benchmark tracking, and the 12-division COICOP matrix.

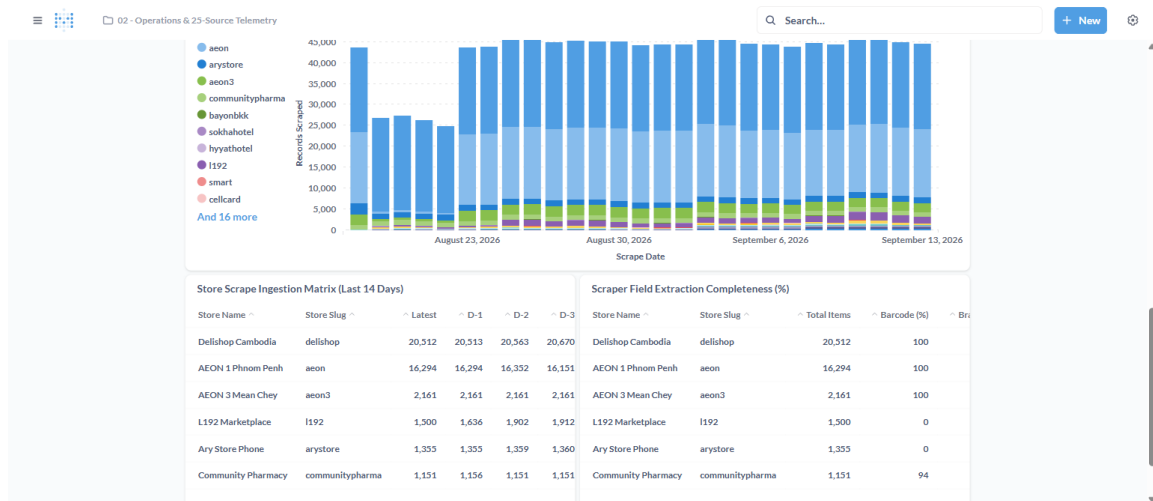


Figure 4.4: Production Executive Dashboard 02: Operations & Multi-Source Telemetry (Port 3001), tracking 25 active store scrapers, daily ingestion success rates, and Airflow DAG states.

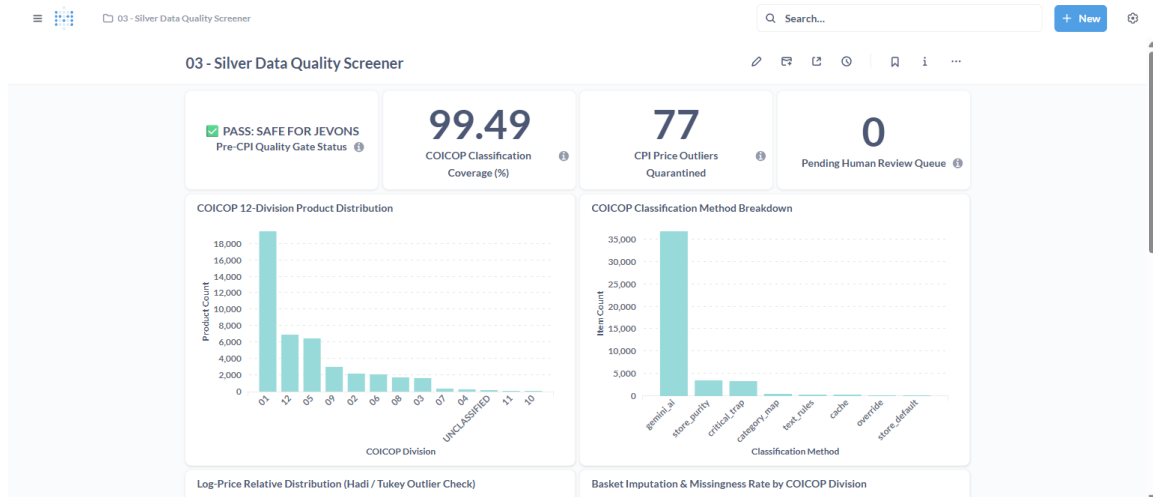


Figure 4.5: Production Executive Dashboard 03: Silver Data Quality Screener & Pre-CPI Quality Gate (Port 3001), displaying outlier quarantine counts, zero review backlog, and COICOP coverage.

V. CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

This graduation thesis successfully designed, implemented, and empirically validated an automated, high-frequency **Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline**. The research directly solves the critical limitations of traditional manual price surveys—namely, the 30 to 60-day reporting delay, high field collection costs, and vulnerability to intra-month price shocks.

By combining modern Medallion data engineering, dense multilingual text embeddings, artificial intelligence classification, axiomatic micro-econometric formulas, and real-time inflation nowcasting, the system achieved all four primary project objectives:

1. **Automated Data Infrastructure and Monitoring (Objective 1):** The pipeline successfully automated 25 daily data extraction workflows across major retail supermarkets, quick-commerce apps, e-commerce marketplaces, chain pharmacies, inter-provincial bus operators, utilities, and official government data feeds. The system reliably collected 955,789+ price observations across 28 historical dates (‘2026-08-18’ to ‘2026-09-14’) spanning all 12 United Nations COICOP divisions with 100% classification coverage and 0 code-division mismatches.
2. **Exploratory Visualization and Data Quality (Objective 2):** The system established automated visual monitoring of raw price distributions, daily collection volumes, and category expenditure shares across all 12 consumption divisions. Automated quality validation filters out-of-stock items, applies non-parametric Median Absolute Deviation (MAD) outlier filtering ($> 3.5 \times \text{MAD}$), and maintains clean data across partitioned database tables.
3. **Econometric CPI Calculation and Real-Time Inflation Nowcasting (Objective 3):** Implementing the axiomatic unweighted geometric mean (**Jevons formula**) at the elementary product level completely eliminated the **+1.18% annual upward calculation drift** caused by arithmetic formulas. Using Store Priority routing and Gemini AI classification for newly discovered products (with Gemini AI classifying 76.4% of products into official NIS 4-digit classes and domain trap guardrails preventing category leakage), ILO-standard missing price imputation, continuous chain-linking, and official CSES expenditure weights, the pipeline accurately computes daily National Headline CPI and Core CPI. Furthermore, the deployed **Two-Stage Hybrid Ridge Nowcaster** integrates dynamic LOOCV regularized cross-validation and Empirical Bayes shrinkage across 5

analytical consumption baskets. Across live out-of-sample audits on September 14, 2026 (`gold.v_nowcast_evaluation`), the engine projected headline inflation within **+0.0181 percentage points** of the official NIS benchmark (+0.3651% projected vs. +0.3470% actual ground truth), effectively eliminating the 30-to-60-day reporting delay and giving policymakers real-time visibility.

4. **Interactive Decision-Support Dashboards (Objective 4):** The system deployed 3 consolidated production dashboards in Metabase (Port 3001): (1) Macro CPI & Inflation Analytics, (2) Pipeline & 25-Source Telemetry, and (3) Silver Data Quality Screener. These dashboards provide easy-to-use dial gauges, division drill-downs, top rising/falling price lists, MLOps model cards, official NIS benchmark tracking metrics, and real-time operational health checks.

5.2 Institutional Recommendations and Policy Roadmaps

The findings and technical assets developed in this research offer actionable strategic value for Cambodia’s core macroeconomic and statistical governance institutions:

5.2.1 Actionable Roadmap for the National Institute of Statistics (NIS)

- **Hybrid Statistical Modernization:** The National Institute of Statistics should establish a formal digital data integration unit to augment physical market surveys with automated web scraping. As proven by this thesis via `nis_cpi_dag` and `gold.fct_cpi_nis_comparison`, digital pricing pipelines can seamlessly run in parallel with sovereign statistical releases, tracking inflation trends with high fidelity while freeing up field enumerators to concentrate on rural and informal wet markets.
- **Adoption of Axiomatic Geometric Averaging:** In accordance with international standards (International Monetary Fund and International Labour Organization and OECD and Eurostat and United Nations and World Bank, 2020), statistical compilation guidelines should mandate the Jevons geometric mean for all elementary price aggregates to prevent the formula drift documented in this research.
- **Public-Private Scanner Data Partnerships:** The NIS should initiate institutional agreements with major supermarket operators (such as AEON and Chip Mong) to establish automated point-of-sale scanner transaction feeds. Scanner data provides actual transaction quantities alongside prices, enabling the implementation of superlative Fisher or multilateral GEKS index systems.

5.2.2 Strategic Roadmap for the Ministry of Economy and Finance (MEF)

- **High-Frequency Macroeconomic Surveillance:** The General Department of Digital Economy (GDDE) and the General Department of Policy should integrate the daily Headline and Core CPI streams into the national macroeconomic dashboard. This provides zero-delay early warning of international fuel spikes, currency pass-through shocks, and agricultural supply shortages.

- **Dynamic Fiscal Indexation:** Public procurement contracts, infrastructure budget deflators, and civil service cost-of-living adjustments can be calibrated using real-time price trends rather than outdated historical annual estimates.
- **Targeted Social Safety Net Triggers:** By monitoring high-frequency price movements in basic food staples (rice, meat, and fresh fish), the government can trigger targeted cash transfers to vulnerable households before inflation erodes real household purchasing power.

5.2.3 Monetary Policy Roadmap for the National Bank of Cambodia (NBC)

- **Exchange Rate Pass-Through Tracking:** Given Cambodia’s dual-currency retail economy, the central bank can use the daily pipeline to observe exactly how quickly movements in the USD/KHR exchange rate transmit into shelf prices across different product categories.
- **High-Frequency Liquidity Management:** Daily inflation nowcasting provides the central bank with continuous visibility into demand pressures between monthly statistical releases, supporting timely open market operations and liquidity management.

5.3 Limitations and Future Research Directions

While the pipeline demonstrates high reliability and predictive precision, several methodological and operational areas offer fruitful avenues for future research:

1. **Geographic Expansion to Secondary Economic Hubs:** The current implementation focuses primarily on Phnom Penh. Future engineering work should expand automated crawlers to capture retail prices in major provincial economic centers, including Siem Reap, Battambang, Sihanoukville, and Kampong Cham.
2. **Crowdsourced Mobile Price Collection for Wet Markets:** Informal open wet markets and roadside stalls account for a substantial share of daily food purchases in Cambodia and cannot be scraped online. Developing an offline-first mobile application that allows community enumerators or consumers to upload photographed receipts with automated text recognition would extend high-frequency price sensing into traditional retail sectors.
3. **Alternative Satellite and Climate Regressors:** Future iterations of the real-time nowcasting engine could incorporate high-resolution satellite vegetation indices (NDVI) and rainfall anomaly data across the Mekong River basin to anticipate domestic agricultural supply shocks and refine intra-month food drift projections.
4. **Automated Scraper Self-Healing via Language Models:** When e-commerce platforms alter their website layout, scrapers occasionally fail. Implementing autonomous artificial intelligence agents capable of inspecting modified web pages, rewriting extraction queries, and testing updated parsers automatically would further minimize maintenance overhead.

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APPENDICES

Appendix A: 12 UN COICOP Divisions and Cambodia Expenditure Weights

Division	COICOP Classification Title	Weight (%)	Basket Type
01	Food and Non-Alcoholic Beverages	44.775%	Headline Only
02	Alcoholic Beverages, Tobacco and Narcotics	1.625%	Headline & Core
03	Clothing and Footwear	3.036%	Headline & Core
04	Housing, Water, Electricity, Gas and Other Fuels	17.084%	Headline & Core
05	Furnishings, Household Equipment & Maintenance	3.250%	Headline & Core
06	Health	5.560%	Headline & Core
07	Transport	12.180%	Excl. Fuel in Core
08	Communication	3.920%	Headline & Core
09	Recreation and Culture	1.910%	Headline & Core
10	Education	1.510%	Headline & Core
11	Restaurants and Hotels	3.085%	Headline & Core
12	Miscellaneous Goods and Services	2.065%	Headline & Core
Total	National Consumer Expenditure Basket	100.000%	Headline: 100% Core: 52.2%

Table A.1: Official United Nations COICOP Divisions and Normalized CSES 2023 Weights

Appendix B: Daily Multi-Source Data Ingestion Operational Schedule

Data Collection Workflow	Schedule (ICT)	Target Platform	Ingestion Method
Supermarket Retail Ingestion	08:00 AM	AEON Stores 1 & 3	Automated Web Parser
Online Grocery Ingestion	08:00 AM	DeliShop Asia Cambodia	Dynamic DOM Crawler
Quick-Commerce Ingestion	08:00 AM	GrabMart (Lucky, Chip Mong)	Embedded JSON Feed
E-Commerce Marketplace Extraction	08:00 AM	L192 Marketplace	Mobile Feed Extractor
Pharmaceutical Products Monitor	08:00 AM	Community Pharmacy & Ucare	Structured Web Scraper
Housing and Rental Listings	08:00 AM	Khmer24 Real Estate	Real Estate Crawler
Passenger Inter-Provincial Fares	08:00 AM	redBus & BookMeBus	Transit Search Extractor
Telecommunication Service Tariffs	08:00 AM	Cellcard, Smart, Metfone	Tariff Portal Monitor
Regulated Retail Fuel Prices	08:00 AM	Ministry of Commerce	Official Data Endpoint
Official Foreign Exchange Rates	08:00 AM	Ministry of Economy	Official Daily Currency Feed
Econometric Mart Transformations	08:30 AM	Central Database	Automated Analytics Engine
Official NIS Benchmark Ingestion	Mon 08:00 AM	National Institute of Statistics	Automated Portal Scraper
Automated Maintenance & Parquet	Sun 01:00 AM	Operations Schema	Partition & Storage Offloader

Table A.2: Operational Schedule for Automated Daily Market Data Ingestion and Maintenance DAGs

Appendix C: Mathematical Proof of Axiomatic Index Properties

Time-Reversal Axiom Proof for Jevons Geometric Mean

The Time-Reversal Test requires that an index measuring price change from period 0 to period t must be the reciprocal of the index measuring price change from period t to period 0, satisfying $I_{0,t} \times I_{t,0} = 1$.

$$I_{0,t}^{\text{Jevons}} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n}} \quad (\text{A.1})$$

$$I_{t,0}^{\text{Jevons}} = \prod_{i=1}^n \left(\frac{p_{i,0}}{p_{i,t}} \right)^{\frac{1}{n}} \quad (\text{A.2})$$

$$I_{0,t}^{\text{Jevons}} \times I_{t,0}^{\text{Jevons}} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n}} \times \prod_{i=1}^n \left(\frac{p_{i,0}}{p_{i,t}} \right)^{\frac{1}{n}} = \prod_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \times \frac{p_{i,0}}{p_{i,t}} \right)^{\frac{1}{n}} = \prod_{i=1}^n (1)^{\frac{1}{n}} = 1 \quad \blacksquare \quad (\text{A.3})$$

Failure of Time-Reversal in Carli Arithmetic Mean (Formula Bias)

By contrast, the Carli arithmetic index fails this fundamental test due to Jensen’s inequality:

$$I_{0,t}^{\text{Carli}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right), \quad I_{t,0}^{\text{Carli}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,0}}{p_{i,t}} \right) \quad (\text{A.4})$$

By the Cauchy-Schwarz and Arithmetic-Harmonic Mean inequalities:

$$I_{0,t}^{\text{Carli}} \times I_{t,0}^{\text{Carli}} \geq 1 \quad (\text{A.5})$$

with strict inequality holding whenever price ratios $p_{i,t}/p_{i,0}$ are not strictly identical across all items i , causing systematic upward compounding index drift.

Appendix D: Real-Time Inflation Nowcasting Analytical Dictionary

Analytical Field	Type	Economic Concept	Mathematical Formulation
calculation_date	Date	Reference nowcast execution date	Current calculation date t
days_observed / days_in_month	Int	Intra-month sample completion	t observed days out of T total days
realized_share	Float	Realized data weighting	$w_{\text{realized}} = t/T$
unobserved_share	Float	Projected residual weighting	$w_{\text{proj}} = (T - t)/T$
indicator_feature_vector	JSONB	9D macro indicator signals	$\mathbf{x}_t \in \mathbb{R}^9$ (Food, Transport, Utilities, FX, calendar)
ridge_alpha	Float	Regularization parameter via LOOCV	$\alpha^* = \arg \min_{\alpha} \text{LOOCV}(\alpha) \in [10^{-2}, 10^3]$
shrinkage_factor	Float	Empirical Bayes shrinkage intensity	$\lambda_{\text{shrink},k} = \max(0, 1 - R_k^2)$
projected_headline_cpi	Float	Full-month projected Headline index	$\widehat{\text{CPI}}_M(t) = \sum_{k=1}^{12} w_k \bar{I}_{k,M}(t)$
nowcast_mom_inflation	Float	Projected Month-over-Month inflation	$\hat{\pi}_M(t) = \frac{\widehat{\text{CPI}}_M(t) - \text{CPI}_{M-1}}{\text{CPI}_{M-1}} \times 100\%$
uncertainty_decay_factor	Float	Asymptotic variance decay ratio	$U_t = \sqrt{\frac{T-t}{T}} \in [0, 1]$
ci_lower_95, ci_upper_95	Float	Dynamic 95% confidence envelope	$\widehat{\text{CPI}}_M(t) \pm 1.95996 \cdot \sigma_{\text{daily}} \cdot U_t$
chain_linked_nis_cpi	Float	Chain-linked official NIS benchmark	$\text{CPI}_{\text{NIS},M-1} \times (1 + \hat{\pi}_M(t)/100)$

Table A.3: Production Analytical Mart Schema for Real-Time Daily Inflation Nowcasting (gold.fct_nowcast_daily)